The Mycelial Web

Trust-Weighted Decentralized Governance
for Post-State Civilization

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Preface

The nation-state is approximately four centuries old. In that span, it has organized humanity's largest cooperative endeavors—and its most catastrophic failures. The Westphalian model assumes territorial sovereignty, monopoly on violence, and hierarchical administration. These assumptions are increasingly inadequate for problems that are global, networked, and ecological in nature.

This book presents Mycelix: a decentralized governance and services platform built on Holochain's agent-centric architecture. With 141+ zomes across 16 cluster DNAs, Mycelix implements a *fractal civic operating system*—a CivOS—that replaces state hierarchy with mycelial networks of mutual aid, earned trust, and commons stewardship.

The central innovation is *trust-weighted governance*: participation is not determined by citizenship, wealth, or age, but by an 8-dimensional sovereign profile measuring epistemic integrity, thermodynamic yield, network resilience, economic velocity, civic participation, stewardship, semantic resonance, and domain competence. This profile gates access to five progressive tiers—Observer, Participant, Citizen, Steward, and Guardian—each with increasing governance capabilities and vote weight.

Mycelix is implemented in approximately 785,000 lines of Rust, with SDKs in TypeScript (226K LOC), Python, and WebAssembly. It is tested by over 8,600 automated tests across all clusters, including sweettest conductor integration tests, property-based tests, and cross-cluster bridge verification.

This book is for system architects, governance theorists, cooperative organizers, and anyone who believes that human coordination can evolve beyond the territorial state.[1]

[1] The philosophical framework underlying these design choices, including the concept of Evolving Resonant Co-creationism and the Seven Primary Harmonies, is elaborated in Stoltz (2026), *Evolving Resonant Co-creationism: A Philosophical Guide to Making It Better, Infinitely.*

Part I

Foundations

Chapter 1

Why Decentralized Governance?

> Neither the life of an individual nor the history of a society can be understood without understanding both.
>
> C. Wright Mills, *The Sociological Imagination* (1959)

1.1 The Crisis of Centralized Coordination

The 21st century confronts humanity with coordination problems that centralized institutions were never designed to solve. Climate change requires cooperation among 195 sovereign states whose incentives are structurally misaligned. Pandemic response requires information sharing across jurisdictions that treat data as sovereign territory. Supply chain resilience requires trust among parties who have no shared legal framework.

These are not merely technical problems. They are *governance* problems—failures of the mechanisms by which humans make collective decisions and enforce collective commitments. The nation-state, optimized for territorial control and industrial economy, is structurally incapable of governing networked, ecological, planetary systems.

1.2 Ostrom and the Commons

Elinor Ostrom's work on common-pool resource management demonstrated that communities can govern shared resources without either state regulation or market privatization. Her eight design principles for long-enduring commons institutions provide the theoretical foundation for Mycelix:

1. **Clearly defined boundaries** — Mycelix implements this through identity verification and the `SovereignProfile`, which defines who can participate and at what level.

2. **Proportional equivalence between benefits and costs** — the trust-weighted participation system ensures that governance power scales with demonstrated commitment (engagement dimension).

3. **Collective-choice arrangements** — the Governance cluster implements proposals, voting, and constitutional mechanisms where all affected parties can participate.

4. **Monitoring** — Holochain's agent-centric architecture provides inherent auditability: every agent's source chain is a tamper-evident log.

5. **Graduated sanctions** — reputation decay (0.998^{days}), slashing ($0.5\times$), and blacklisting (threshold < 0.05) implement progressive enforcement.

6. **Conflict-resolution mechanisms** — the Justice domain in the Civic cluster provides structured dispute resolution.

7. **Minimal recognition of rights to organize** — Mycelix operates on Holochain, which requires no permission from any authority to deploy.

8. **Nested enterprises** — the fractal cluster architecture (16 clusters, each with multiple domain zomes) implements precisely this principle of nested, polycentric governance.

1.3 Beyond the Nation-State

The nation-state model assumes several properties that Mycelix explicitly rejects:

Territorial sovereignty Mycelix governance is not tied to geographic territory. A water stewardship council in the Commons cluster governs a watershed, not a political boundary.

Monopoly on legitimate violence Mycelix replaces coercion with reputation consequences. The only "force" is reduced governance participation for those who lose community trust.

Hierarchical administration Mycelix is fractal: every cluster is self-governing, connected to others through typed bridge dispatch rather than bureaucratic hierarchy.

Citizenship by birth or naturalization Mycelix membership is earned through demonstrated multi-dimensional civic engagement across eight sovereign dimensions. The `ConsciousnessTier` system replaces birthright citizenship with progressive, earned governance capacity.

1.4 The Mycelial Metaphor

Mycelium—the vegetative body of fungi—forms vast underground networks that connect individual organisms into cooperative webs. A single mycelial network can span hectares,

sharing nutrients between trees, signaling threats, and decomposing dead matter into fertile soil.

Mycelix adopts this metaphor deliberately:

- **No central node**: Like mycelium, Mycelix has no server, no headquarters, no single point of failure.
- **Nutrient sharing**: The Finance cluster's three currencies (SAP, TEND, MYCEL) flow through the network like nutrients through hyphae.
- **Decomposition**: Demurrage and currency composting ensure that hoarded value returns to the commons, just as mycelium decomposes dead organic matter.
- **Signal propagation**: Cross-cluster bridge dispatch propagates governance signals across domain boundaries, like chemical signals propagating through mycelial networks.

1.5 The Great Filter of Co-Creative Wisdom

The cosmos is silent. Despite hundreds of billions of stars in our galaxy alone, many harboring Earth-like planets, the universe presents no evidence of other technological civilizations. The Fermi Paradox has traditionally been explained by two categories of "Great Filter": either the emergence of intelligent life is extraordinarily improbable (the filter is behind us), or technological civilizations inevitably destroy themselves through nuclear war, engineered pandemics, or misaligned artificial intelligence (the filter is ahead of us). A third possibility is more subtle and more final: civilizations fail not from spectacular self-destruction, but from *coordination failure*—the inability to develop ethical and social technologies commensurate with their computational and physical power. Under this model, civilizations that fail the test do not end in galaxy-illuminating explosions. They stagnate, fragment, and devolve. They lose the capacity to distinguish signal from noise, to trust one another, and to coordinate collective action. To the rest of the universe, they simply go silent. The Great Silence is not the sound of explosions but the sound of a billion civilizations failing to mature.

Humanity currently stands at the threshold of this filter. Our technological capacity—exponentially accelerated by artificial intelligence—vastly outpaces our collective wisdom and ethical maturity. We possess the power to reshape our planet and our biology, but our social operating systems remain rooted in territorial sovereignty, zero-sum competition, and hierarchical administration. AI, deployed to optimize these legacy systems rather than to transform them, acts as an accelerant: it collapses shared epistemic foundations through disinformation, erodes trust through surveillance and algorithmic bias, and paralyzes governance through coordination failure. The result is not a single cataclysm but a slow drowning in complexity and incoherence.

Mycelix is a concrete architectural response to the Great Filter. If the failure mode is a collapse of trust, sense-making, and coordination, then the remedy must be an architecture that systematically generates and scales trust through decentralized identity, verifiable claims, and wise collective governance. The fractal cluster design, the trust-weighted participation system, and the Epistemic Charter described in the chapters that follow are not

merely technical conveniences—they are the social technologies that the Great Filter hypothesis identifies as the missing prerequisite for civilizational survival. A complete treatment of this argument appears in Stoltz (2025), *The Great Filter of Co-Creative Wisdom: A Socio-Technical Solution to the Fermi Paradox* [12].

1.6 Design Philosophy: Eight Harmonies

Mycelix is guided by the Eight Harmonies framework, which provides ethical constraints on system design:

1. **Radical Transparency**: All governance actions are auditable on agent source chains.
2. **Inclusive Participation**: Trust-weighted gating ensures progressive access rather than binary inclusion/exclusion.
3. **Ecological Reciprocity**: The Climate and Energy clusters embed environmental stewardship into economic activity.
4. **Intergenerational Care**: Constitutional mechanisms in the Governance cluster can encode protections for future generations.
5. **Distributed Power**: No single agent can accumulate unchecked governance authority; Guardian tier requires high scores across all eight sovereign dimensions.
6. **Creative Expression**: The Music and Education clusters support human flourishing beyond mere governance.
7. **Emergent Wisdom**: Trust-weighted participation trusts that collective intelligence emerges from well-structured governance access.
8. **Sacred Stillness**: The system includes mechanisms for pause, reflection, and deliberation before irreversible actions.

Chapter 2

Holochain Architecture

> Integrity doesn't require consensus.
>
> Arthur Brock and Eric Harris-Braun,
> Holochain Whitepaper (2018)

2.1 Agent-Centric, Not Data-Centric

Blockchain architectures are data-centric: they maintain a single, globally consistent ledger that all participants must agree upon. This design choice entails fundamental trade-offs—high energy consumption (proof-of-work), limited throughput (global consensus bottleneck), and a conceptual model where data exists independently of the agents who create it.

Holochain inverts this model. Each agent maintains their own *source chain*: a hash-linked, append-only log of their actions. Data integrity is local and immediate—an agent signs every entry they commit. There is no global ledger and no global consensus.

2.1.1 Source Chains

An agent's source chain is a sequence of headers, each containing:

- The hash of the previous header (chain linkage)
- The agent's public key signature
- The hash of the entry data (if any)
- A sequence number and timestamp

This structure provides tamper evidence without requiring any other agent's participation. If an agent modifies a historical entry, the chain of hashes breaks, and any peer performing validation will detect the tampering.

2.1.2 The DHT: Shared Awareness Without Global State

While source chains provide local integrity, the *distributed hash table* (DHT) provides shared awareness. When an agent commits an entry, a copy is sent to a neighborhood of peers determined by the entry's hash. These peers *validate* the entry against the DNA's validation rules before storing it.

Key properties of Holochain's DHT:

- **Sharded storage**: Each agent stores only a neighborhood of the total data, proportional to the number of agents.
- **Redundancy**: Each entry is stored by multiple peers (configurable), providing resilience against node departure.
- **Eventual consistency**: The DHT converges to consistency as gossip propagates, but never requires synchronous global agreement.
- **Intrinsic validation**: Every peer independently validates entries against the DNA's rules. There is no "trusted validator" role.

2.2 DNA as Application Definition

A Holochain DNA defines the rules of a particular application space. It consists of:

Integrity zomes define entry types, link types, and validation rules. These are the "constitution" of the application: immutable, deterministic, and enforced by every peer.

Coordinator zomes implement application logic—CRUD operations, queries, and cross-zome calls. These can be updated without changing the DNA hash.

This separation is crucial for Mycelix. The integrity zomes define the *rules of governance*—what constitutes a valid proposal, a valid vote, a valid identity credential. The coordinator zomes implement the *mechanisms of governance*—how proposals are submitted, how votes are tallied, how credentials are issued.

2.3 hApp Architecture

A Holochain application (hApp) bundles one or more DNAs into roles. Each role is instantiated as a *cell* on the conductor—the Holochain runtime. Cells within the same conductor can communicate via `call(CallTargetCell, ...)`.

Mycelix uses a *unified hApp* architecture: all 16 cluster DNAs are bundled into a single hApp with defined roles. This enables cross-cluster communication via `CallTargetCell::OtherRole`, avoiding the complexity of inter-hApp signaling.

Listing 2.1: Unified hApp manifest (simplified)

```yaml
---
manifest_version: "1"
name: mycelix-unified
roles:
  - name: commons
    dna: mycelix-commons.dna
  - name: civic
    dna: mycelix-civic.dna
  - name: identity
    dna: mycelix-identity.dna
  - name: governance
    dna: mycelix-governance.dna
  - name: finance
    dna: mycelix-finance.dna
  # ... 11 more cluster roles
```

2.4 Why Not Blockchain?

The choice of Holochain over blockchain (Ethereum, Solana, etc.) is deliberate and follows from Mycelix's design requirements:

Property	**Blockchain**	**Holochain**
Consensus	Global (all nodes agree)	Local (source chain + DHT validation)
Scaling	O(n) per transaction	O(1) per agent action
Data model	Global state (ledger)	Agent-centric (source chains)
Energy	PoW: enormous; PoS: moderate	Minimal (no mining)
Governance	Token-weighted (plutocratic)	Identity-weighted (meritocratic)
Privacy	Pseudonymous but public	Private source chains + selective sharing
Offline	Impossible (requires consensus)	Possible (local source chain)

The most critical difference for governance applications is the consensus model. Blockchain governance is inherently plutocratic: those with more tokens have more votes. Holochain's agent-centric model allows Mycelix to implement trust-weighted governance where influence derives from identity, reputation, community trust, and engagement rather than wealth.

Chapter 3

The Fractal Cluster Design

> A complex system that works is invariably found to have evolved from a simple system that worked.
>
> John Gall, *Systemantics* (1975)

3.1 Why 16 Clusters?

Mycelix organizes its 133+ zomes into 16 cluster DNAs. This is neither arbitrary nor aspirational—it emerges from two constraints:

1. **Holochain's cell model**: Each DNA is instantiated as a cell. Cross-zome calls within a cell (`CallTargetCell::Local`) are fast and synchronous. Cross-cell calls (`CallTargetCell::OtherRole`) require message passing.

2. **Domain cohesion**: Zomes that frequently interact should share a DNA. A housing governance vote that checks property records should not require a cross-cluster call.

The 16 clusters group by domain affinity:

Cluster	Domain	Zomes	Status
mycelix-commons	Property, Housing, Care, Mutual Aid, Water, Food, Transport	39	Built
mycelix-civic	Justice, Emergency, Media, Resonance Feed	18	Built
mycelix-hearth	Kinship, Gratitude, Care, Autonomy, Decisions	12	Built
mycelix-identity	DID, MFA, Trust Credentials, Recovery, Web-of-Trust	13	Built
mycelix-governance	Proposals, Voting, DKG, Councils, Constitution	7	Built
mycelix-finance	SAP/TEND/MYCEL Payments, Treasury, Staking	8	Built
mycelix-health	FHIR, CDS, Telehealth, Trials, Insurance	15	Built
mycelix-knowledge	Claims, Graph, Query, Inference, Factcheck	8	Built
mycelix-personal	Identity Vault, Health Vault, Credential Wallet	4	Built

Cluster	Domain	Zomes	Status
mycelix-attribution	Dependency Registry, Usage Receipts, Reciprocity	3	Built
mycelix-mail	PQC-Encrypted Decentralized Email	12	Built
mycelix-supplychain	Provenance Tracking	8	Built
mycelix-marketplace	Arbitration	8	Built
mycelix-energy	Projects, Investments, Grid, Regenerative	5	Built
mycelix-climate	Carbon Credits, Projects	3	Built
mycelix-space	Orbital Objects, Debris Bounties, Traffic Control	5	Built

Additional clusters (Music, Education, DeSci, Core) bring specialized capabilities for cultural production, learning, and research.

3.2 Anatomy of a Cluster

Each cluster follows a consistent internal architecture:

Integrity zomes (one per domain) define entry types and validation rules. For example, `housing_integrity` defines `HousingUnit`, `Membership`, and `MaintenanceRequest` entry types.

Coordinator zomes (one per domain) implement CRUD operations, queries, and domain logic. For example, `housing_units` provides `create_unit`, `list_units`, `transfer_unit`.

Bridge zome (one per cluster) handles cross-cluster dispatch. It receives typed requests from other clusters and routes them to the appropriate domain zome via local calls.

Shared types crate (`mycelix-bridge-entry-types`) defines DHT entry types shared across clusters.

3.3 The Fractal Pattern

The architecture is fractal in a precise sense: the same governance pattern repeats at every scale.

- **Zome level**: Each coordinator zome checks the caller's trust credential before performing governance-sensitive operations.
- **Cluster level**: Each cluster's bridge zome enforces cluster-wide policies (rate limiting, allowlists) before routing requests.
- **System level**: The unified hApp manifest defines which clusters can communicate, and the bridge-common crate provides type-safe dispatch across all clusters.

This fractal structure means that a water stewardship circle, a city-wide housing cooperative, and the global Mycelix network all use the same governance primitives: trust profiles, tiered access, typed bridge dispatch.

3.4 Shared Infrastructure: bridge-common

The `crates/mycelix-bridge-common` crate (295+ tests) provides the shared infrastructure that all clusters depend on:

- `SovereignProfile` — the 8D sovereign profile (epistemic integrity, thermodynamic yield, network resilience, economic velocity, civic participation, stewardship, semantic resonance, domain competence) profile (detailed in Chapter 4)
- `ConsciousnessTier` — the 5-tier enum (Observer through Guardian)
- `ConsciousnessCredential` — time-limited, BLAKE3-committed credentials
- `BridgeDomain` — typed domain routing (12 variants)
- `CommonsZome` / `CivicZome` — typed zome routing (38 + 15 variants)
- Rate limiting, allowlists, and reputation primitives

Part II

Trust-Weighted Governance

Chapter 4

The 8D Sovereign Profile

> Who should govern? is the wrong question. The right question is: How can we organize institutions so that bad or incompetent rulers can be prevented from doing too much damage?
>
> Karl Popper, *The Open Society and Its Enemies* (1945)

4.1 Terminology and Scope

The codebase uses `SovereignProfile` and `CivicTier` for the types described in this chapter, replacing the earlier `ConsciousnessProfile` (4D) with an eight-dimensional model. The eight dimensions are each measured directly from a primary source cluster, not from self-reports or financial proxies:

- D_0: **Epistemic Integrity** — truth-telling track record (knowledge cluster)
- D_1: **Thermodynamic Yield** — physical energy contribution (energy/grid)
- D_2: **Network Resilience** — node uptime and bandwidth (mesh-time)
- D_3: **Economic Velocity** — anti-hoarding TEND circulation (finance)
- D_4: **Civic Participation** — voting, jury duty, proposals (governance)
- D_5: **Stewardship & Care** — verified commons labor (attribution)
- D_6: **Semantic Resonance** — federated learning Hamming similarity (core-FL)
- D_7: **Domain Competence** — peer-verified expertise with Ebbinghaus decay (craft)

These names reflect Mycelix's architecture as a consciousness-coupled governance platform. In the standalone Mycelix context—where no Symthaea engine is connected—these eight dimensions function as *earned trust-weighted participation gates*. The mechanism is straightforward: agents accumulate scores across eight physically-grounded dimensions, and the composite determines governance capability. The `gate_civic()` function provides backward-compatible gating: it attempts native 8D evaluation first, falling back to the legacy

4D `ConsciousnessProfile` with automatic conversion.

A reader who is skeptical of computational consciousness can safely read the sovereign profile as a "multi-dimensional earned trust capacity profile." The eight dimensions, the tier system, the sigmoid vote weighting, and the credential mechanism are all defined by their engineering properties, not by any metaphysical commitment. The system's value lies in its measurable governance outcomes: Sybil resistance, progressive access, reputation accountability, and community-weighted trust. These properties hold regardless of one's position on machine consciousness.

4.2 Motivation: Beyond Binary Access Control

Traditional access control is binary: you either have permission or you do not. Role-based access control (RBAC) adds granularity but remains fundamentally categorical. A user is either "admin" or "editor" or "viewer."

This model is inadequate for governance. Governance participation should be *progressive*—increasing with demonstrated trustworthiness, commitment, and community standing. A newcomer should be able to observe and learn before participating. A long-standing community member should have more governance weight than a newcomer, but not infinite weight.

Mycelix replaces binary access control with an *8-dimensional sovereign profile* that produces a continuous governance capability score. No single dimension's weight can exceed 50% of the composite—a constitutional invariant that prevents single-axis capture.

4.3 The Eight Dimensions

Each dimension is a continuous value in $[0, 1]$, measured from a primary source cluster:

Definition 4.1 (SovereignProfile)**.** *A trust profile* $\mathbf{P} = (D_0, D_1, \ldots, D_7) \in [0, 1]^8$ *where:*

- D_0 *(Epistemic Integrity)* $\in [0, 1]$*: Truth-telling track record from validated claims in the knowledge cluster. Saturates at 50 validated claims.*
- D_1 *(Thermodynamic Yield)* $\in [0, 1]$*: Physical energy contribution from verified energy/grid records. Saturates at 30 verified records.*
- D_2 *(Network Resilience)* $\in [0, 1]$*: Node uptime and bandwidth provision from mesh-time logs. Saturates at 720 time anchors.*
- D_3 *(Economic Velocity)* $\in [0, 1]$*: Anti-hoarding compliance from TEND mutual credit exchanges. Saturates at 50 TEND exchanges.*
- D_4 *(Civic Participation)* $\in [0, 1]$*: Voting, jury duty, and proposal activity from governance records. Saturates at 20 governance actions.*
- D_5 *(Stewardship & Care)* $\in [0, 1]$*: Verified commons labor from attribution/reciprocity pledges. Saturates at 30 pledges.*

- D_6 *(Semantic Resonance)* $\in [0, 1]$*: Community value alignment via Hamming similarity between agent and consensus training hypervectors in federated learning.*

- D_7 *(Domain Competence)* $\in [0, 1]$*: Peer-verified expertise from living credentials with Ebbinghaus decay. Saturates at 3 living credentials.*

Constitutional Decay. All dimensions decay exponentially without engagement: $S_{\text{decayed}} = S_{\text{raw}} \cdot e^{-\lambda \cdot \Delta t}$, where λ is community-configurable within constitutional bounds ($\lambda_{\text{min}} = 0.001$/day, half-life 693 days; $\lambda_{\text{max}} = 0.020$/day, half-life 35 days). Lambda = 0 is constitutionally impossible, preventing permanent oligarchy.

Privacy. The sovereign profile never leaves the participant's device. Governance proofs use zero-knowledge STARK proofs (∼1ms prove, ∼0.3ms verify, ∼7.5KB proof) that demonstrate threshold compliance without revealing raw dimensions.

Legacy Compatibility. The earlier 4D `ConsciousnessProfile` (Identity, Reputation, Community, Engagement) remains in the codebase for backward compatibility. The `gate_civic()` function evaluates the native 8D profile first, falling back to 4D with automatic conversion: Identity $\rightarrow D_0 + D_2$, Reputation $\rightarrow D_3 + D_5$, Community $\rightarrow D_4 + D_6$, Engagement $\rightarrow D_1 + D_7$.

4.3.1 Identity Dimension

The identity dimension measures how strongly an agent has verified their identity. It maps directly from the MFA (Multi-Factor Authentication) assurance level defined in the Identity cluster:

- **Anonymous** ($I = 0.0$): No identity verification. The agent has a Holochain public key but nothing else.

- **Basic** ($I = 0.25$): Single-factor authentication (e.g., email verification).

- **Verified** ($I = 0.50$): Multi-factor authentication (e.g., email + TOTP).

- **HighlyAssured** ($I = 0.75$): Strong multi-factor (e.g., hardware security key + biometric).

- **Critical** ($I = 1.0$): Maximum assurance, including identity proofing by trusted community members.

This dimension provides Sybil resistance: creating fake identities becomes progressively more expensive as higher assurance levels require more verification.

4.3.2 Reputation Dimension

The reputation dimension aggregates an agent's behavioral history across all Mycelix clusters. It uses exponential decay with a 30-day half-life:

$$R(t) = \sum_i w_i \cdot r_i \cdot \left(\frac{1}{2}\right)^{(t-t_i)/30} \tag{4.1}$$

where r_i is the reputation event value, t_i is the event timestamp (in days), w_i is the source weight, and t is the current time.

Key properties:

- **Temporal decay**: Reputation fades with time, preventing ancient good behavior from conferring permanent privilege.
- **Multi-source**: Reputation events come from all clusters, weighted by domain relevance.
- **Slashing**: Severe misbehavior triggers a 0.5× multiplier on accumulated reputation.
- **Blacklist threshold**: Agents whose reputation falls below 0.05 are blacklisted from governance.
- **Restoration**: Blacklisted agents can restore reputation through 100 verified positive interactions.

4.3.3 Community Dimension

The community dimension measures peer trust—how much the agent's community vouches for them. Trust attestations are weighted by the attestor's own governance tier:

$$C = \frac{\sum_j \tau_j \cdot w(\text{tier}_j)}{\sum_j w(\text{tier}_j)} \tag{4.2}$$

where τ_j is attestor j's trust rating and $w(\text{tier}_j)$ is the weight assigned to their tier. This creates a virtuous cycle: attestations from higher-tier community members carry more weight, incentivizing genuine community building rather than attestation farming.

4.3.4 Engagement Dimension

The engagement dimension measures domain-specific participation. Unlike the other three dimensions (which are global), engagement is computed locally by each cluster's bridge zome. This means an agent can have high engagement in the Health cluster but low engagement in the Finance cluster.

For Symthaea-connected agents, the engagement dimension maps from the unified consciousness score (C_{unified}), creating a bridge between Symthaea's consciousness research and Mycelix governance participation.

4.4 Combined Score

The combined score is a weighted average:

$$S = 0.25 \cdot I + 0.25 \cdot R + 0.30 \cdot C + 0.20 \cdot E \tag{4.3}$$

The weights reflect a deliberate design choice: community trust (30%) is weighted highest because Mycelix is fundamentally a system of *mutual recognition*. Identity and reputation (25% each) provide the foundation. Engagement (20%) ensures active participation matters but cannot substitute for the other dimensions.

4.5 Continuous Vote Weight

Rather than using hard tier thresholds for vote weight, Mycelix employs a sigmoid function centered at the Citizen threshold (0.4):

$$W(S) = \frac{W_{\max}}{1 + e^{-(S-0.4)/T}} \tag{4.4}$$

where $W_{\max} = 10{,}000$ basis points and $T = 0.05$ is the temperature parameter. This provides:

- Smooth transitions (no "cliff" at tier boundaries)
- Noise tolerance (small score fluctuations don't cause dramatic weight changes)
- Configurable strictness (lower T = sharper transition)

Chapter 5

Five Governance Tiers

5.1 Tier Definitions

Definition 5.1 (ConsciousnessTier)**.** *Five governance tiers, derived from the combined trust score S:*

1. ***Observer*** *($S < 0.3$): Read-only access. Can browse public data and observe governance proceedings.*
2. ***Participant*** *($S \geq 0.3$): Can submit basic proposals and comment on governance actions. Vote weight: 5,000 bp (50%).*
3. ***Citizen*** *($S \geq 0.4$): Full voting rights on standard governance actions. Vote weight: 7,500 bp (75%).*
4. ***Steward*** *($S \geq 0.6$): Can initiate constitutional amendments, manage council membership, and access cross-cluster governance. Vote weight: 10,000 bp (100%).*
5. ***Guardian*** *($S \geq 0.8$): Emergency governance powers including circuit-breaker activation, emergency fund release, and threat response coordination. Vote weight: 10,000 bp (100%).*

5.2 Hysteresis: Preventing Tier Oscillation

A naive threshold system would cause agents near tier boundaries to oscillate rapidly as their scores fluctuate from measurement noise. Mycelix implements hysteresis with a margin of ± 0.05:

- **Promotion** requires crossing threshold $+ 0.05$
- **Demotion** requires dropping below threshold $- 0.05$

An agent at Citizen tier (threshold 0.4) must reach 0.45 to promote to Steward, but only drops to Citizen if they fall below 0.55 ($0.6 - 0.05$). This creates a stability band around each tier boundary.

5.3 Governance Requirements

Each governance action specifies requirements as a `GovernanceRequirement` struct:

```rust
pub struct GovernanceRequirement {
    pub min_tier: ConsciousnessTier,
    pub min_identity: Option<f64>,
    pub min_community: Option<f64>,
    pub min_reputation: Option<f64>,
    pub min_engagement: Option<f64>,
}
```

This allows fine-grained control. A constitutional amendment might require:

- Minimum tier: Steward ($S \geq 0.6$)
- Minimum identity: 0.75 (HighlyAssured)
- Minimum community: 0.5 (strong peer attestation)

5.4 Credentials: Time-Limited Trust

Trust profiles are not queried in real-time (which would require cross-cluster calls during governance actions). Instead, each agent's bridge zome issues a `ConsciousnessCredential`—a time-limited (24-hour default) credential containing the profile, derived tier, and a BLAKE3 trajectory commitment.

The credential is stored on the agent's source chain and presented to governance zomes as proof of tier. This design:

- Eliminates cross-cluster calls during governance actions
- Provides a 24-hour validity window (stale data is bounded)
- Includes a BLAKE3 hash of the agent's behavioral trajectory, preventing credential transfer between agents

5.5 Bootstrap Mode

New communities face a bootstrapping problem: no one has reputation or community trust yet. Mycelix handles this with bootstrap mode:

- Communities with fewer agents than `BOOTSTRAP_COMMUNITY_THRESHOLD` operate under relaxed gating
- Minimum identity of `BOOTSTRAP_MIN_IDENTITY` is still required (Sybil resistance is never relaxed)

- Bootstrap credentials have a separate TTL (`BOOTSTRAP_TTL_US`)
- As the community grows past the threshold, standard gating activates automatically

Chapter 6

Bridge-Common: Type-Safe Cross-Cluster Dispatch

6.1 The Problem: Fragile String Matching

In early Mycelix prototypes, cross-cluster dispatch used string matching:

```
if query_type.contains("transfer") {
    call(housing_transfer_zome, ...)
} else if query_type.contains("membership") {
    call(housing_membership_zome, ...)
}
```

This was fragile: a query type of `"transfer_membership"` would match both branches. Case sensitivity caused bugs. New zomes required updating string patterns across multiple clusters.

6.2 Typed Routing

The `bridge-common` crate replaces string matching with strongly typed enums:

```
pub enum BridgeDomain {
    Property, Housing, Care, Mutualaid,
    Water, Food, Transport, Support,
    Space, Justice, Emergency, Media,
}

pub enum CommonsZome {
    PropertyRegistry, PropertyTransfer,
    PropertyDisputes, PropertyCommons,
    HousingUnits, HousingMembership,
    // ... 38 total variants
}
```

Each zome variant maps to a canonical snake_case string matching the actual Holochain zome name. The routing function `resolve_commons_zome` performs case-insensitive domain parsing via `BridgeDomain::from_str_loose`, which handles variations like "mutualaid", "mutual_aid", and "mutual-aid".

6.3 Cross-Cluster Roles

Communication between clusters uses Holochain's `CallTargetCell::OtherRole` mechanism. The `CrossClusterRole` enum defines the available roles:

```
pub enum CrossClusterRole {
    Commons,
    Civic,
    Identity,
    Governance,
    Finance,
    // ... all 16 cluster roles
}
```

Each cross-cluster call specifies the target role, target zome, and function name. The bridge-common crate provides allowlists that restrict which source-destination pairs are permitted, preventing unauthorized cross-cluster access.

6.4 Rate Limiting and Allowlists

Bridge dispatch includes two security mechanisms:

Allowlists specify which source clusters can call which destination zomes. For example, only the Governance cluster can call the Identity cluster's trust credential verification functions.

Rate limiting prevents any single agent from overwhelming a cluster's bridge with requests. Limits are configurable per domain and per agent tier.

6.5 The 295+ Test Suite

The bridge-common crate is one of the most heavily tested components in Mycelix, with 295+ tests covering:

- Profile construction and validation (NaN handling, clamping)
- Tier derivation and hysteresis
- Credential issuance and expiry
- Domain routing (case insensitivity, unknown domains)

- Vote weight computation (sigmoid properties, edge cases)
- Reputation decay, slashing, and blacklisting
- Property-based tests (tier monotonicity, score bounds)

Chapter 7

Collective Intelligence

> None of us is as smart as all of us.
>
> Ken Blanchard

7.1 Beyond Individual Trust

The preceding chapters describe how individual agents earn trust through the 4D profile. But governance is a *collective* phenomenon—the quality of a community's decisions depends not only on the trustworthiness of individual participants but on the coherence of the group as a whole. Mycelix addresses this through three mechanisms: collective Phi, fragile consensus detection, and the Epistemic Charter.

7.2 Collective Phi: Group Consciousness Vectors

When Symthaea is connected, each agent produces a consciousness state characterized by six measurable dimensions: integrated information (Φ), workspace activation (GWT), attention coherence, meta-cognitive depth, emotional valence, and social-model accuracy. These dimensions form a 6-dimensional *agent consciousness vector*:

$$\mathbf{c}_i = (\Phi_i,\ W_i,\ A_i,\ M_i,\ V_i,\ S_i) \tag{7.1}$$

For any pair of agents i, j, the pairwise coherence is the cosine similarity of their consciousness vectors:

$$\kappa_{ij} = \frac{\mathbf{c}_i \cdot \mathbf{c}_j}{\|\mathbf{c}_i\|\ \|\mathbf{c}_j\|} \tag{7.2}$$

The *collective Phi* of a governance quorum Q is defined as:

$$\Phi_Q = \frac{2}{|Q|(|Q|-1)} \sum_{i<j\in Q} \kappa_{ij} \tag{7.3}$$

This metric quantifies how aligned a deliberating group's cognitive states are. A collective Phi above 0.7 indicates coherent deliberation; below 0.3 suggests fragmentation that may produce low-quality decisions.

7.3 Fragile Consensus Detection

Not all consensus is genuine. A vote that passes with 51% in favor may reflect deep division rather than collective agreement. Mycelix detects fragile consensus by analyzing:

Vote margin The gap between the winning and losing tallies. Margins below a configurable threshold trigger extended deliberation periods.

Tier distribution A proposal that passes only because Guardian-tier agents outvoted many Citizen-tier agents may reflect elite capture rather than community consensus.

Coherence gap When collective Phi is low but the vote margin is high, the consensus may be fragile—agents voted the same way for different reasons and may defect when circumstances change.

Temporal stability If pre-vote sentiment polling shows rapid opinion changes, the consensus may not be durable.

When fragile consensus is detected, the Governance cluster can automatically:

1. Extend the voting period
2. Require a higher approval threshold
3. Flag the decision for review after a cooling-off period
4. Recommend facilitated dialogue before re-vote

7.4 Reputation System

The reputation system provides the temporal dimension of collective intelligence. It ensures that trust is earned, maintained, and can be lost:

Exponential decay Reputation decays at 0.998^{days}, yielding a half-life of approximately 346 days. This prevents ancient contributions from conferring permanent influence while allowing sustained good behavior to accumulate meaningful trust.

Slashing Severe violations—data falsification, Sybil attacks, governance manipulation—trigger a $0.5\times$ multiplicative slash on accumulated reputation. The slash is immediate and recorded on the agent's source chain.

Blacklisting Agents whose reputation falls below 0.05 lose all governance participation. This threshold is deliberately low: it takes sustained or severe misbehavior to reach it.

Restoration Blacklisted agents can restore reputation through 100 verified positive interactions across at least two clusters. This ensures that restoration requires genuine community re-engagement, not merely waiting out a timer.

7.5 The Epistemic Charter (E-N-M Classification)

Governance decisions vary in their epistemic character. A proposal to fund a bridge repair is fundamentally different from a proposal to change the community's constitutional values. The Epistemic Charter provides a three-axis classification for governance claims:

E (Empirical) Scored E0–E4, from Subjective (E0: personal testimony) to Publicly Reproducible (E4: independently verified by multiple observers). Higher E levels require stronger evidence and more verifiers.

N (Normative) Scored N0–N3, from Personal preference (N0) to Axiomatic principles (N3: foundational values that define the community). Higher N levels require broader consensus and longer deliberation.

M (Materiality) Scored M0–M3, from Ephemeral (M0: temporary, easily reversible) to Foundational (M3: permanent, defines the community's structure). Higher M levels require higher-tier approval and constitutional protections.

A governance action's E-N-M classification determines:

- The minimum tier required to propose and vote
- The evidence standard for supporting claims
- The approval threshold (simple majority for low-M, supermajority for high-M)
- The deliberation period (hours for M0, weeks for M3)

For example, a claim that "the water pH in the eastern watershed is 6.8" is classified E3-N0-M1 (empirically measurable, no normative content, moderate materiality). A proposal that "the community shall adopt a policy of restorative justice" is E0-N3-M3 (not empirically testable, deeply normative, foundational). The governance requirements differ accordingly.

Part III

Core Clusters

Chapter 8

Identity

8.1 Overview

The Identity cluster is the foundation of Mycelix. Every other cluster depends on it for agent verification, trust credentials, and the identity dimension of the trust profile. It contains 13 zomes across 11 domains plus 1 bridge and 1 shared:

1. **did_registry** — DID (Decentralized Identifier) management. Each agent registers a `did:mycelix:<pubkey>` identifier.

2. **mfa** — Multi-Factor Authentication with configurable assurance levels (Anonymous through Critical).

3. **trust_credential** — Peer-to-peer trust attestations that feed the community dimension.

4. **verifiable_credential** — W3C Verifiable Credential issuance and verification.

5. **credential_schema** — Schema registry for credential types.

6. **recovery** — Social recovery mechanism (threshold of trusted peers can restore lost keys).

7. **revocation** — Credential and key revocation with DHT propagation.

8. **education** — Educational credential management.

9. **name-registry** — Human-readable name registration and resolution.

10. **web-of-trust** — Transitive trust computation across the network.

11. **bridge** — Cross-cluster identity verification dispatch.

8.2 DID Architecture

Mycelix uses a method-specific DID scheme: `did:mycelix:<pubkey>`. Each DID document is stored on the agent's source chain and replicated to the DHT. The document contains:

- The agent's public key (Ed25519)
- Optional post-quantum key (ML-KEM-768, for future-proofing)
- Service endpoints (which clusters the agent participates in)
- Authentication methods (linked MFA assurance level)
- Delegation chains (sub-passports for authorized devices)

8.3 Multi-Factor Authentication

The MFA zome implements five assurance levels, each requiring progressively stronger verification:

Level	Identity Score	Requirements
Anonymous	0.00	Holochain key only
Basic	0.25	Email or phone verification
Verified	0.50	Two independent factors
HighlyAssured	0.75	Hardware key + biometric
Critical	1.00	In-person proofing by trusted peers

8.4 Web of Trust

The web-of-trust zome computes transitive trust across the network. If Alice trusts Bob and Bob trusts Carol, Alice has indirect trust in Carol, discounted by a configurable decay factor per hop. This enables:

- Discovery of trustworthy agents outside one's direct community
- Sybil detection (clusters of mutually-attesting fake identities become visible as isolated subgraphs)
- Community dimension computation (aggregated trust from the web)

8.5 Social Recovery

The recovery zome implements m-of-n social recovery: an agent designates n trusted peers, and any m of them can collectively authorize key rotation. This eliminates the single point of failure of traditional key management while preserving decentralization.

Chapter 9

Governance

9.1 Overview

The Governance cluster implements the decision-making infrastructure. Its 7 zomes (plus 1 bridge) provide a complete governance toolkit:

1. **proposals** — Proposal creation, amendment, and lifecycle management.
2. **voting** — Multiple voting mechanisms (simple majority, ranked choice, quadratic, conviction).
3. **threshold-signing** — Distributed Key Generation (DKG) for threshold signatures on governance decisions.
4. **councils** — Council formation, membership management, and delegation.
5. **constitution** — Constitutional document management with amendment procedures.
6. **execution** — Automated execution of approved governance decisions.
7. **jurisdiction** — Jurisdictional boundary management for nested governance.
8. **bridge** — Cross-cluster governance dispatch.

9.2 Proposal Lifecycle

1. **Draft**: Any Participant+ can create a proposal draft.
2. **Discussion**: The community discusses and suggests amendments.
3. **Voting**: Once discussion period ends, voting opens. The voting mechanism depends on the proposal type (configured per jurisdiction).
4. **Resolution**: Votes are tallied. Trust-weighted vote totals determine the outcome.
5. **Execution**: Approved proposals trigger automated execution via the execution zome, which dispatches the appropriate cross-cluster calls.

9.3 Distributed Key Generation

The threshold-signing zome implements a DKG protocol that produces shared keys without any single party knowing the complete private key. This enables:

- Multi-signature treasury operations (Finance cluster integration)
- Constitutional amendment signing (requires Guardian-tier quorum)
- Emergency response authorization (distributed circuit-breaker)

9.4 Constitutional Governance

The constitution zome manages the foundational rules of a Mycelix community. Constitutional documents are versioned, and amendments require:

- Proposer: Steward tier minimum
- Identity dimension: ≥ 0.75 (HighlyAssured)
- Community dimension: ≥ 0.5
- Supermajority vote (configurable, default 67%)
- Mandatory discussion period (configurable, default 14 days)

Chapter 10

Finance

10.1 Three-Currency Economics

The Finance cluster implements a three-currency system designed to serve different economic functions:

SAP (Sapience) A stable unit of account pegged to a basket of real goods and services. Used for pricing, contracts, and accounting. Cannot be created ex nihilo—minted only through verified economic activity. Subject to demurrage, with compost redistributed to commons pools as a maintenance flow rather than a hoarding subsidy.

TEND (Tenderness) A mutual credit currency for care work, community service, and non-market contributions. Zero-sum by construction: every positive balance is matched by an equal negative balance elsewhere in the network. TEND does not implement demurrage.

MYCEL (Mycelium) A commons currency representing collective stewardship. Used for treasury operations, public goods funding, and inter-community exchange.

10.2 Cluster Zomes

1. **currency-mint** — Currency creation and supply management.
2. **payments** — Transaction processing for all three currencies.
3. **tend** — TEND-specific mutual credit operations.
4. **treasury** — Trust-gated treasury management.
5. **staking** — MYCEL staking for governance weight.
6. **recognition** — Non-monetary recognition and gratitude tokens.
7. **price-oracle** — Decentralized price discovery for SAP basket.
8. **bridge** — Cross-cluster financial operations.

10.3 Trust-Gated Treasury

Treasury operations are gated by governance tier:

- **View treasury balance**: Observer+
- **Propose expenditure**: Citizen+
- **Approve expenditure**: Steward+ (threshold signature required)
- **Emergency fund release**: Guardian (DKG multi-sig)

10.4 Demurrage and Composting

SAP implements demurrage—a periodic reduction in idle balance that reflects the carrying cost of the real basket backing the currency and discourages passive hoarding. This is inspired by Silvio Gesell's Freigeld ("free money") concept and implemented as:

$$B(t) = B(t_0) \cdot e^{-\lambda(t-t_0)} \tag{10.1}$$

where λ is the demurrage rate. The "composted" value (the amount lost to demurrage) is routed to commons pools, completing the metabolic cycle.

TEND, by contrast, is mutual credit rather than a composting currency. Its stability comes from bounded credit lines, reciprocity, and zero-sum settlement, not from timed balance decay.

Currency composting is the explicit mechanism by which retired or excess currency units are removed from circulation. It is idempotent: composting an already-composted unit is a no-op, and the total supply is always zero-sum (minted − composted = circulating).

10.5 Oracle and Price Discovery

The price-oracle zome implements decentralized price feeds for the SAP basket. Multiple oracle agents submit price attestations, and an accuracy-weighted trimmed median is used as the canonical price. Oracle submissions require Citizen tier minimum, and when there is a prior consensus the oracle compares new consensus against the previous stored median to detect volatility and trigger counter-cyclical TEND limit expansion. contract includes protections against:

- Division by zero (NaN prices)
- Stale prices (TTL enforcement)
- Manipulation (outlier detection, reputation-weighted aggregation)

Chapter 11

Commons

11.1 The Largest Cluster

The Commons cluster, with 39 zomes across 9 domains (plus mesh-time and resource-mesh), is the largest cluster in Mycelix. It implements the practical infrastructure of daily community life.

11.2 Domain Overview

11.2.1 Property (4 zomes)

- **property_registry**: Land and building registration with provenance tracking.
- **property_transfer**: Ownership transfer with trust-gated approval.
- **property_disputes**: Structured dispute resolution for property conflicts.
- **property_commons**: Community land trust management.

11.2.2 Housing (6 zomes)

Housing cooperatives are first-class citizens in Mycelix:

- **housing_units**: Unit registry and availability tracking.
- **housing_membership**: Cooperative membership management.
- **housing_finances**: Shared housing financial operations.
- **housing_maintenance**: Maintenance request and scheduling.
- **housing_clt**: Community Land Trust governance.
- **housing_governance**: Housing-specific governance (distinct from the system-level Governance cluster).

11.2.3 Care (5 zomes)

- **care_timebank**: Time-banking for care work, integrated with TEND currency.
- **care_circles**: Care circle formation and coordination.
- **care_matching**: Matching care needs with available caregivers.
- **care_plans**: Structured care planning for individuals and families.
- **care_credentials**: Caregiver credential verification.

11.2.4 Mutual Aid (7 zomes)

- **mutualaid_needs**: Need posting and categorization.
- **mutualaid_circles**: Mutual aid circle management.
- **mutualaid_governance**: Circle-level governance.
- **mutualaid_pools**: Resource pooling for collective needs.
- **mutualaid_requests**: Request processing and fulfillment tracking.
- **mutualaid_resources**: Resource inventory and availability.
- **mutualaid_timebank**: Mutual aid time banking.

11.2.5 Water (5 zomes)

- **water_flow**: Water flow monitoring and allocation.
- **water_purity**: Water quality testing and reporting.
- **water_capture**: Rainwater harvesting coordination.
- **water_steward**: Watershed stewardship governance.
- **water_wisdom**: Traditional and scientific water knowledge.

11.2.6 Food (4 zomes)

- **food_production**: Community garden and farm management.
- **food_distribution**: Food distribution coordination.
- **food_preservation**: Preservation knowledge and scheduling.
- **food_knowledge**: Seed libraries, recipes, growing guides.

11.2.7 Transport (3 zomes)

- **transport_routes**: Community transport route planning.
- **transport_sharing**: Vehicle and ride sharing coordination.
- **transport_impact**: Environmental impact tracking.

11.2.8 Cross-Cutting: Mesh-Time and Resource-Mesh

Two cross-cutting zomes serve the entire Commons cluster:

- **mesh-time**: Coordinated time-keeping across domains, enabling temporal queries ("what water quality readings existed when this food was distributed?").
- **resource-mesh**: Cross-domain resource allocation and tracking, enabling a community garden to request water allocation from the water domain.

11.3 Testing: 5,276 Tests

The Commons cluster is tested with 5,276 automated tests—the highest count of any single cluster. These include unit tests for each zome, integration tests for cross-zome operations, and sweettest conductor tests for full Holochain runtime validation.

Chapter 12

Hearth: Family and Community

> The family is one of nature's masterpieces.
>
> George Santayana, *The Life of Reason* (1905)

12.1 Why a Family Cluster?

Governance frameworks typically model individuals as atomistic rational actors. But humans live in families, households, and kinship networks. Care work, child rearing, elder support, and collective decision-making within families are among the most consequential governance activities in any community—yet they are invisible to most civic infrastructure.

The Hearth cluster (12 zomes, 1,023 tests) brings family and community relationships into Mycelix as first-class governance primitives.

12.2 Domain Overview

12.2.1 Kinship Networks (2 zomes)

The kinship zomes model family and chosen-family relationships:

- **kinship_registry**: Registration of kinship bonds (biological, adoptive, chosen-family) with mutual consent. Both parties must confirm the relationship before it is committed to the DHT.

- **kinship_governance**: Family-level governance—shared decision-making weighted by kinship role and trust. Parents, elders, and designated guardians receive contextual governance weight within the kinship network.

12.2.2 Gratitude Exchange (1 zome)

The gratitude zome implements a non-monetary recognition system for care work, emotional support, and community contribution. Gratitude tokens:

- Are non-transferable (cannot be hoarded or traded)
- Feed the community dimension of the sender's and receiver's trust profiles
- Decay with time (30-day half-life), preventing gratitude inflation
- Are visible to the kinship network but private to outsiders

12.2.3 Care Coordination (1 zome)

The care zome coordinates within-family care responsibilities:

- Scheduling shared care duties (childcare, elder care, disability support)
- Matching care needs with available family members
- Integration with the Commons cluster's care timebank for extended community care

12.2.4 Autonomy Tracking (1 zome)

A distinctive feature of Hearth: the autonomy zome tracks the developing autonomy of minors and dependents. As a child grows, their governance capability within the family (and eventually the broader community) increases progressively. This mirrors the trust tier system at the family scale:

- Young children: Observer (family decisions are visible to them)
- Older children: Participant (can express preferences)
- Adolescents: Citizen (vote weight in family governance)
- Adults: Full autonomy (Steward/Guardian eligible)

12.2.5 Collective Decisions (1 zome)

Family-level collective decisions use trust-weighted voting where weights are determined by kinship role, age, and demonstrated engagement with the family's governance. A family deciding where to live, how to allocate a shared resource, or how to respond to a community need uses the same governance primitives as the system-level Governance cluster.

12.2.6 Story Archiving (1 zome)

The stories zome preserves family narratives, oral histories, and cultural knowledge. Stories are:

- Stored on the author's source chain (tamper-evident)
- Linked to kinship relationships ("grandmother's recipe")
- Optionally shared with the broader community or the Knowledge cluster
- Tagged with epistemic classification (E0-N2-M2 for cultural knowledge)

12.2.7 Milestones, Rhythms, Emergency, and Resources (4 zomes)

Milestones Track family events (births, graduations, ceremonies) as governance-relevant markers that can trigger tier transitions.

Rhythms Model recurring family patterns (meal schedules, care rotations) that the mesh-time system can synchronize across households.

Emergency Family-level emergency coordination, integrated with the Civic cluster's emergency system for escalation.

Resources Shared family resource tracking (tools, vehicles, spaces), with lending and return workflows.

12.2.8 Bridge (1 zome)

The Hearth bridge dispatches cross-cluster requests—most commonly to Identity (for kinship credential verification), Commons (for care coordination), and Governance (for family-level proposal escalation to community governance).

12.3 Trust-Weighted Family Governance

Family governance in Hearth is trust-weighted but with different dimension priorities than system-level governance. The community dimension (peer trust) carries the highest weight within a kinship network, because family trust is fundamentally relational rather than transactional.

The 1,023 tests cover kinship lifecycle (creation, modification, dissolution), gratitude exchange mechanics, autonomy progression, and cross-cluster bridge dispatch. 48 sweettest integration tests verify full Holochain runtime behavior, including 9 trust-gating tests that validate tier requirements for family governance actions.

Chapter 13

Personal and Attribution

13.1 Personal: Private Vaults

The Personal cluster (4 zomes including bridge, 20 tests) implements private data management:

13.1.1 Identity Vault

The identity vault stores the agent's most sensitive identity artifacts:

- Recovery seed phrases (encrypted, never DHT-replicated)
- Biometric templates (stored locally, used for MFA verification)
- Delegation keys (sub-passports for authorized devices)
- Identity dimension history (track MFA level changes over time)

The vault is strictly source-chain-only: entries are never committed to the DHT, ensuring that even a compromised DHT neighborhood cannot access the agent's core identity material.

13.1.2 Health Vault

The health vault stores personal health data under patient control:

- Medical records (FHIR-formatted, from the Health cluster)
- Prescription history
- Genetic data (if voluntarily submitted)
- Mental health assessments

Data sharing is opt-in and granular: an agent can share lab results with their care circle (via the Hearth cluster) without exposing their full medical history.

13.1.3 Credential Wallet

The credential wallet manages the agent's verifiable credentials:

- W3C Verifiable Credentials from the Identity cluster
- Educational credentials from the EduNet cluster
- Care credentials from the Commons cluster
- Governance tier credentials (ConsciousnessCredentials)

The wallet presents credentials selectively—an agent proving they are Citizen tier need not reveal their identity dimension score, only the derived tier.

13.2 Attribution: Tracking Reciprocity

The Attribution cluster (3 zomes, 17 tests) provides cross-cutting infrastructure for tracking who contributed what to whom:

Dependency Registry Tracks dependency relationships between creative works, software, datasets, and community resources. When an educational curriculum uses a Knowledge cluster claim, the dependency is recorded.

Usage Receipts Immutable records of resource usage. When the Music cluster plays a track, a usage receipt is created on the listener's source chain and replicated to the artist's DHT neighborhood.

Reciprocity Engine Computes net reciprocity scores between agents and communities. An agent who consumes many community resources but contributes little will see their engagement dimension decrease. An agent who contributes generously will see it increase.

Attribution integrates with Finance (TEND payments for usage), Music (artist royalties), Education (curriculum attribution), and Knowledge (citation tracking).

Part IV

Domain Clusters

Chapter 14

Health

14.1 15-Zome Architecture

The Health cluster is Mycelix's most medically comprehensive component, with 15 active zomes organized in two tiers:

Tier 1 (MVP):

1. Patient records management
2. Provider registry and credentialing
3. Appointment scheduling
4. Prescription management
5. Lab results tracking
6. Referral coordination
7. Billing and insurance claims

Tier 2 (Advanced):

1. **FHIR** — HL7 FHIR resource modeling for healthcare interoperability.
2. **CDS** — Clinical Decision Support with tier-gated recommendation generation.
3. **Telehealth** — Decentralized telemedicine session management.
4. **Trials** — Clinical trial coordination and data collection.
5. **Insurance** — Decentralized mutual health insurance pools.
6. **Nutrition** — Nutritional planning and dietary management.
7. **Mental health** — Mental health assessment and tracking.
8. **Public health** — Epidemiological surveillance and reporting.

14.2 FHIR Interoperability

The FHIR (Fast Healthcare Interoperability Resources) zome implements HL7 FHIR R4 resource types as Holochain entries. This enables:

- Data exchange with existing healthcare systems
- Standardized patient records across Mycelix communities
- Regulatory compliance with healthcare data standards

14.3 Privacy Architecture

Health data requires the strongest privacy protections. The Health cluster implements:

- Patient-controlled data sharing (source chain privacy)
- Encrypted DHT entries for sensitive records
- Tier-gated access (provider credentialing required)
- Audit trails on all data access

14.4 Clinical Decision Support

The CDS (Clinical Decision Support) zome is tier-gated at two levels:

1. **Provider access**: Only agents with verified healthcare credentials (from the Identity cluster's education and verifiable credential zomes) can query the CDS system.
2. **Recommendation generation**: CDS recommendations are tagged with the governance tier of the requesting provider and the evidence level of the recommendation. A Participant-tier provider receives only well-established guidelines, while a Steward-tier provider can access experimental protocols and off-label usage data.

This tiered access prevents information overload for less-experienced providers while ensuring that experienced practitioners have access to the full knowledge base.

14.5 Mutual Health Insurance

The Insurance zome implements decentralized mutual health insurance pools:

- Community members contribute to a shared pool (denominated in TEND or MYCEL)
- Claims are submitted with medical evidence (from the patient records zome) and adjudicated by a panel of Citizen+ tier pool members

- Large claims require Steward-level approval and DKG multi-signature from the Governance cluster
- Pool governance follows the same trust-weighted model as system-level governance

This model eliminates the profit motive of commercial insurance while maintaining the risk-pooling benefits.

Chapter 15

Knowledge

15.1 Epistemic Infrastructure

The Knowledge cluster implements decentralized knowledge verification—an "epistemic mesh" where claims are verified through multiple independent pathways:

1. **claims** — Structured claim registration with provenance metadata.
2. **graph** — Knowledge graph construction with typed relationships.
3. **query** — Semantic query processing across the knowledge graph.
4. **inference** — Logical inference engine for deriving new claims from existing ones.
5. **factcheck** — Multi-source fact verification with confidence scoring.
6. **markets_integration** — Prediction market integration for epistemic value discovery.
7. **bridge** — Cross-cluster knowledge dispatch.

15.2 Claim Lifecycle

1. **Registration**: An agent registers a claim with supporting evidence. The claim entry includes: statement, evidence links, confidence, domain tags.
2. **Verification**: Independent agents verify or dispute the claim. Verification quality is weighted by the verifier's governance tier and domain engagement.
3. **Inference**: The inference engine derives implications from verified claims, extending the knowledge graph automatically.
4. **Factchecking**: The factcheck zome aggregates verification results into a confidence score.
5. **Market pricing**: Prediction markets assign economic value to claim accuracy, creating incentives for truthful reporting.

15.3 Knowledge Graph

The graph zome maintains a typed knowledge graph where nodes are claims and edges are relationships (supports, contradicts, extends, specializes). The graph is distributed across the DHT, with each neighborhood validating the relationships in its shard.

Query processing supports:

- Path queries ("what claims support this conclusion?")
- Subgraph extraction ("all claims in the climate domain")
- Confidence propagation ("what is the transitive confidence in this claim?")

15.4 Prediction Markets

The markets_integration zome connects knowledge claims to economic incentives:

1. A Citizen+ agent creates a prediction market for a specific, verifiable claim with a defined resolution date.
2. Participants stake MYCEL on the outcome (true/false/probability).
3. Market prices reflect the collective's confidence in the claim.
4. Upon resolution, correct predictors are rewarded proportionally, and their reputation dimension increases.
5. Incorrect predictions reduce reputation, creating accountability.

The prediction market mechanism provides a "skin in the game" complement to the expert verification pathway. Claims that have both high expert verification scores and high market confidence are the most robustly validated.

15.5 Integration Patterns

The Knowledge cluster integrates with other clusters through several patterns:

Health → Knowledge Medical claims from clinical trials feed the knowledge graph, enabling meta-analysis across decentralized research sites.

Climate → Knowledge Carbon credit verification claims are registered in the knowledge graph and can be independently verified through satellite observation data from the Space cluster.

Supply Chain → Knowledge Provenance claims from the supply chain are cross-referenced with production data from the Commons cluster's food and water domains.

Education → Knowledge Educational content references knowledge graph claims, ensuring that curricula are backed by verified knowledge.

Chapter 16

Space and Climate

16.1 Space: Orbital Stewardship

The Space cluster addresses a growing governance challenge: who manages orbital space? With 5 zomes plus an orbital mechanics library, it implements:

1. **orbital_objects** — Satellite and debris tracking registry.
2. **observations** — Decentralized observation reporting from ground stations and amateur astronomers.
3. **conjunctions** — Collision probability computation and warning dissemination.
4. **debris_bounties** — Economic incentives for debris removal, funded through MYCEL commons currency.
5. **traffic_control** — Decentralized orbital traffic management protocols.

The orbital mechanics library provides Keplerian propagation, conjunction assessment, and maneuver planning—the computational backbone for governance decisions about orbital resource allocation.

16.2 Climate: Carbon and Beyond

The Climate cluster manages environmental stewardship:

1. **carbon** — Carbon credit registry with tier-gated verification. Credits are issued for verified carbon sequestration and can be traded or retired.
2. **projects** — Climate project registration, monitoring, and impact reporting.
3. **bridge** — Cross-cluster climate data dispatch (e.g., carbon impact of supply chain operations).

16.3 Integration: Space-Climate Nexus

Space and climate governance intersect in satellite-based Earth observation: the Space cluster's observation network can verify climate project claims (deforestation monitoring, agricultural carbon sequestration) through the Knowledge cluster's factcheck pipeline.

16.4 Debris Economics

The debris_bounties zome creates economic incentives for orbital cleanup:

1. Debris objects are cataloged in the orbital_objects registry with estimated removal difficulty and collision risk.
2. The community allocates MYCEL bounties proportional to the object's collision risk (computed by the conjunctions zome).
3. Removal operators claim bounties by submitting observation evidence that the debris has been deorbited or moved to a graveyard orbit.
4. Claims are verified through the Knowledge cluster's factchecking pipeline using independent observations from multiple ground stations.

This mechanism applies the same trust-weighted governance to space stewardship: bounty creation requires Citizen+ tier, bounty approval requires Steward+, and emergency collision avoidance can be triggered by Guardian-tier agents.

16.5 Carbon Credit Lifecycle

The carbon zome implements a full carbon credit lifecycle:

1. **Project registration**: A climate project (reforestation, renewable energy, soil carbon) is registered with baseline measurements.
2. **Monitoring**: Periodic measurement reports are submitted with evidence links (satellite data, sensor readings, third-party audits).
3. **Credit issuance**: Verified sequestration generates carbon credits, recorded as Holochain entries on the project operator's source chain.
4. **Trading**: Credits can be transferred between agents, retired (permanently removed from circulation), or used to offset emissions.
5. **Retirement**: Retired credits are composted (similar to currency composting in the Finance cluster), ensuring they cannot be double-counted.

All operations are tier-gated: credit issuance requires Citizen+ with verified engagement in the climate domain, and credit retirement is auditable through the source chain.

Chapter 17

Music and Education

17.1 Music: Attribution-First Royalties

The Music cluster (4 zomes plus 14 support crates) implements a royalty-free-by-default, attribution-required music platform:

1. **catalog** — Music catalog management with provenance tracking.
2. **plays** — Play count tracking and listener analytics.
3. **balances** — Artist balance management (TEND-denominated).
4. **trust** — Artist verification and anti-plagiarism.

The Attribution cluster (`mycelix-attribution`) provides the cross-cutting infrastructure: dependency registry, usage receipts, and reciprocity tracking.

17.2 Education: Curriculum and Credentials

The Education cluster (10 zomes, `mycelix-edunet`) implements:

- Curriculum design and sharing
- Learner progress tracking
- Peer assessment with tier-gated credentialing
- Educational credential issuance (integrated with Identity cluster)
- Knowledge graph integration for learning path recommendation

Chapter 18

Civic Infrastructure

> Justice is what love looks like in public.
>
> Cornel West

18.1 17 Zomes for Civic Life

The Civic cluster (18 zomes including bridge, 2,273 tests) implements three domains of civic infrastructure: restorative justice, emergency coordination, and decentralized media. Together they address the functions that nation-states perform through courts, emergency services, and regulated press—reimagined for a decentralized, trust-weighted governance model.

18.2 Restorative Justice

The Justice domain explicitly rejects the punitive model of criminal records, incarceration, and permanent stigma. Instead, it implements restorative justice—a process-oriented approach that seeks to repair harm rather than punish offenders.

18.2.1 Process, Not Punishment

The justice zomes implement a structured conflict resolution lifecycle:

1. **Incident report**: Any Participant+ agent can file a structured incident report describing harm experienced. Reports are stored on the reporter's source chain and shared with a mediator pool.

2. **Mediator assignment**: Trained mediators (Citizen+ tier with justice domain engagement) are assigned based on availability, domain expertise, and absence of conflict of interest.

3. **Circle process**: The mediator convenes a restorative circle including the affected parties, community supporters, and (optionally) the person who caused harm. The process follows structured dialogue protocols.

4. **Agreement**: The circle produces a restorative agreement—a set of commitments by the responsible party (restitution, community service, behavioral change) with a timeline and verification criteria.

5. **Monitoring**: Agreement compliance is tracked through the governance system. Completion feeds the reputation dimension positively; non-compliance triggers graduated sanctions (reputation decay, tier reduction).

18.2.2 No Permanent Records

A critical design decision: completed restorative processes are *archived* rather than permanently attached to an agent's identity. After a configurable period (default: 1 year post-completion), the incident hash is removed from the agent's public profile links. The source chain entry remains (Holochain source chains are append-only), but it is no longer discoverable through DHT queries. This implements the "right to rehabilitation" in code.

18.3 Emergency Coordination

The Emergency domain provides infrastructure for disaster response, community crises, and urgent coordination:

Incidents Structured incident reporting with severity levels, geolocation, and resource requirements. High-severity incidents trigger automatic cross-cluster notifications.

Triage Resource triage using a priority queue weighted by severity, affected population, and available resources. Triage decisions require Citizen+ tier and are auditable.

Shelters Shelter registration, capacity tracking, and occupancy management. Integrated with the Commons cluster's housing domain for emergency housing allocation.

Communications Emergency broadcast system using the bridge dispatch to reach all clusters simultaneously. Guardian-tier agents can trigger network-wide emergency alerts. When the mesh radio bridge is available (feature `mesh`), emergency alerts propagate via LoRa radio even when internet connectivity is lost.

18.4 Decentralized Media

The Media domain implements a decentralized publishing and curation platform:

Publication Content publishing with provenance tracking. Every published piece is signed and timestamped on the author's source chain, providing tamper-evident authorship.

Curation Community-driven content curation using trust-weighted recommendations. Higher-tier curators' selections carry more weight, but all Participant+ agents can curate.

Fact-checking Integration with the Knowledge cluster's factcheck pipeline. Media claims can be tagged for verification, and the verification status is displayed alongside the content.

Attribution Integration with the Attribution cluster for proper crediting of sources, quotes, and derivative works.

Resonance Feed A trust-weighted content feed that surfaces content based on community trust signals rather than engagement metrics. This explicitly rejects the attention-economy model that rewards outrage and sensationalism.

18.5 Testing: 2,273 Tests

The Civic cluster's 2,273 tests cover justice process lifecycle, emergency coordination workflows, media publication and curation, and cross-cluster bridge dispatch. 17 sweettest integration tests verify full conductor-level behavior, including cross-domain interactions (e.g., an emergency incident triggering a justice mediation process).

Chapter 19

Mail, Supply Chain, and Marketplace

19.1 Mail: PQC-Encrypted Decentralized Email

The Mail cluster (12 zomes) implements a fully decentralized email system with post-quantum cryptographic security:

Key exchange ML-KEM-768 (NIST FIPS 203) key encapsulation for establishing shared secrets between correspondents.

Message encryption AES-256-GCM symmetric encryption using the ML-KEM-derived shared secret.

Mailbox management Each agent maintains a private mailbox on their source chain. Incoming messages are encrypted to the recipient's public key and stored in their DHT neighborhood.

Address book Decentralized address resolution using the Identity cluster's name-registry zome.

Attachments Large attachments are chunked, encrypted, and distributed across the DHT with content-addressed hashes.

Spam prevention Trust-weighted message filtering: messages from agents below Observer tier are quarantined by default. Community trust dimension modulates filter sensitivity.

The 12 zomes cover the full email lifecycle: compose, encrypt, send, receive, decrypt, reply, forward, search, folder management, contact management, attachment handling, and bridge dispatch. All message content is end-to-end encrypted; even DHT validators cannot read message bodies.

19.2 Supply Chain: Provenance Tracking

The Supply Chain cluster (8 zomes) implements end-to-end provenance tracking for physical goods:

Inventory Product registration with unique identifiers, batch tracking, and location history.

Logistics Shipment management, route optimization, and delivery confirmation with cryptographic proof-of-delivery.

Procurement Purchase order management with trust-gated supplier approval (Citizen+ tier required for high-value procurement).

Verification Product authenticity verification using source chain provenance—each handler's signature provides a link in the chain of custody.

Trust Supplier reputation tracking integrated with the system-wide reputation dimension.

Claims Dispute resolution for supply chain disagreements, integrated with the Civic cluster's justice domain.

Payments Supply chain payment processing using SAP (for contracted prices) and TEND (for informal exchanges).

Bridge Cross-cluster dispatch for supply chain queries from other clusters (e.g., Food domain querying provenance of ingredients).

19.3 Marketplace: Decentralized Commerce with Arbitration

The Marketplace cluster (8 zomes) provides a decentralized commerce platform:

- Product and service listing with tier-gated seller verification
- Escrow-based transactions using the Finance cluster's payment infrastructure
- Buyer and seller reputation tracking
- Structured arbitration process for disputes, escalating to the Civic cluster's justice domain when marketplace-level resolution fails
- Trust-weighted product reviews (higher-tier reviewers' assessments carry more weight)
- Integration with the Supply Chain cluster for physical goods provenance verification

Chapter 20

Energy and DeSci

20.1 Energy: Community Power

The Energy cluster (5 zomes) manages community-scale energy production, investment, and distribution:

Projects Energy project registration—solar installations, wind turbines, micro-hydro, community batteries—with capacity, generation forecasts, and maintenance schedules.

Investments Trust-gated community investment in energy projects. Investment proposals require Citizen+ tier. Large investments (above a configurable threshold) require Steward+ approval and DKG multi-signature from the Governance cluster.

Regenerative Regenerative energy practices tracking—soil carbon sequestration from biochar, forest management, and agricultural energy recovery.

Grid Community microgrid management: energy balance tracking, peer-to-peer energy trading (denominated in MYCEL), and demand response coordination.

Bridge Cross-cluster dispatch for energy data queries from Climate (carbon offsets from renewable generation), Finance (energy investment returns), and Commons (resource-mesh integration).

20.2 DeSci: Decentralized Science

The DeSci cluster (`mycelix-desci`) takes a different architectural approach from other Mycelix clusters: it implements a REST API (using Actix-web) rather than Holochain zomes. This design choice reflects the DeSci cluster's role as a bridge between the Mycelix governance network and the existing scientific research ecosystem.

20.2.1 REST API Architecture

The DeSci REST API provides:

- Research project registration and metadata management
- Dataset registration with provenance tracking
- Peer review coordination with trust-weighted reviewer assignment
- Reproducibility tracking (linking research claims to data, code, and computational environment)
- Integration with the Knowledge cluster for claim verification
- Open access publishing with attribution tracking

20.2.2 141 Integration Tests

The DeSci cluster is validated by 141 integration tests that exercise the full REST API surface, including:

- Project lifecycle (create, update, archive)
- Dataset upload, versioning, and access control
- Review assignment, submission, and aggregation
- Cross-cluster queries (Knowledge graph, Attribution)
- Authentication and authorization (trust profile verification)

20.2.3 Federated Learning Integration

DeSci integrates with the `mycelix-core` federated learning framework for privacy-preserving collaborative research. Research institutions can participate in multi-site studies without sharing raw patient data—only trust-weighted gradient updates are exchanged through the 0TML pipeline.

Chapter 21

PQC-Encrypted Decentralized Mail

> The right of the people to be secure in their persons, houses, papers, and effects, against unreasonable searches and seizures, shall not be violated.
>
> Fourth Amendment, U.S. Constitution

21.1 Why Decentralized Mail Requires Post-Quantum Cryptography

Centralized email is the single largest vector for state surveillance and corporate data harvesting. The `mycelix-mail` cluster (12 zomes, approximately 100,000 lines of Rust) replaces the entire SMTP/IMAP stack with a Holochain-native messaging system where *no central mail server exists* and every message is encrypted with lattice-based post-quantum primitives.

The cryptographic foundation is ML-KEM-768 (NIST FIPS 203), a module-lattice key encapsulation mechanism that remains secure against known quantum algorithms. Forward secrecy is achieved without a dedicated key server: each correspondence pair performs a double-ratchet key exchange mediated entirely through DHT entries on the participants' source chains.

21.2 Zome Architecture

The 12 zomes partition the full email lifecycle into orthogonal concerns:

Zome	Responsibility
`compose`	Message drafting, template management
`encrypt`	ML-KEM encapsulation, AES-256-GCM bulk encryption

Zome	Responsibility
`send`	DHT neighborhood publication, delivery receipts
`receive`	Incoming message polling, decryption delegation
`decrypt`	Private-key operations via LAIR keystore
`threads`	Conversation threading, in-reply-to resolution
`attachments`	Content-addressed chunking, encrypted distribution
`contacts`	Address book, Identity cluster name-registry integration
`folders`	Inbox/sent/archive/custom folder management
`search`	Encrypted search over ciphertext (trapdoor-based)
`spam`	Trust-weighted spam filtering (see Section 21.5)
`bridge`	Cross-cluster dispatch for mail queries

21.3 Forward Secrecy Without a Key Server

Classical forward secrecy relies on ephemeral Diffie–Hellman exchanges coordinated by a trusted server. Mycelix eliminates the server by storing ephemeral public keys as *private DHT entries* in the recipient's neighborhood. The sender retrieves the latest ephemeral key, encapsulates a shared secret via ML-KEM, and discards the ephemeral pair after use. The result is per-message forward secrecy: compromise of a long-term private key does not retroactively decrypt past messages.

21.4 Encrypted Search Over Ciphertext

Full-text search is critical for usability but appears incompatible with end-to-end encryption. Mycelix implements *trapdoor-based search*: the sender generates encrypted keyword indices (trapdoors) alongside the ciphertext. The recipient's search zome matches query trapdoors against stored trapdoors without decrypting message bodies. This provides substring-level search with zero plaintext exposure to DHT validators.

21.5 Spam Filtering and Trust Integration

Spam filtering leverages the 8D sovereign profile (Chapter 4):

- Messages from agents below Observer tier are quarantined by default.

- Community trust dimension modulates filter sensitivity—agents vouched for by the recipient's web-of-trust bypass quarantine.
- Reputation dimension provides a decay-based spam cost: repeated spam reports slash sender reputation ($0.5\times$), eventually triggering blacklist (< 0.05).
- Attachment size limits scale with sender tier (Observer: 1 MB, Guardian: 100 MB).

21.6 Thread Management and Attachments

Conversations are modeled as directed acyclic graphs of DHT entries. The `threads` zome maintains link tags for `in-reply-to`, `references`, and `forward-of` relationships. Large attachments are split into 256 KB chunks, individually encrypted, and distributed across the sender's DHT neighborhood with content-addressed BLAKE3 hashes. The recipient reassembles chunks using the manifest entry attached to the parent message.

Chapter 22

Supply Chain Provenance

> Trust, but verify.
>
> Russian proverb, popularized by Ronald Reagan

22.1 End-to-End Provenance Without Consensus Bottleneck

Traditional supply chain blockchains suffer from the consensus bottleneck: every transaction must be globally ordered by miners or validators, limiting throughput and creating single points of failure. The `mycelix-supplychain` cluster (8 zomes) uses Holochain's agent-centric DHT to record provenance *locally* on each handler's source chain while the DHT's validation rules ensure global consistency without global consensus.

Each handler in the chain of custody—farmer, processor, shipper, retailer—signs a `ProvenanceEntry` on their source chain. Downstream handlers verify the upstream signature before accepting custody. The result is a cryptographically linked chain of custody that is tamper-evident, auditable, and operates at the throughput of individual agents rather than global block intervals.

22.2 Zome Architecture

Zome	Responsibility
`inventory`	Product registration, batch tracking, location history
`logistics`	Shipment management, route optimization, delivery confirmation

Zome	Responsibility
`procurement`	Purchase orders, tier-gated supplier approval (Citizen+)
`verification`	Authenticity verification via source chain provenance
`trust`	Supplier reputation, integrated with system-wide reputation
`claims`	Dispute resolution, Civic cluster justice integration
`payments`	SAP (contracted) and TEND (informal) payment processing
`bridge`	Cross-cluster dispatch (e.g., Food domain querying provenance)

22.3 Trust-Weighted Supplier Verification

Supplier verification uses the 8D sovereign profile (Chapter 4) with domain-specific extensions:

1. **Identity**: Verified business registration via the Identity cluster's credential wallet.
2. **Reputation**: Delivery reliability score (on-time rate, defect rate, dispute history) with temporal decay (0.998^{days}).
3. **Community**: Peer endorsements from existing supply chain participants, weighted by endorser tier.
4. **Engagement**: Volume and frequency of successful transactions in the past 90 days.

High-value procurement (above a configurable threshold) requires Citizen-tier or above for both buyer and supplier, enforced by the consciousness gating layer in `bridge-common`.

22.4 IoT Sensor Integration for Cold Chain Monitoring

The logistics zome supports `SensorReading` entries for temperature, humidity, GPS coordinates, and shock/vibration events. IoT gateways publish signed sensor readings to the handler's source chain at configurable intervals. Validation rules enforce:

- Temperature readings within product-specific acceptable ranges.
- GPS coordinates consistent with the declared shipping route.
- Timestamp monotonicity (no backdated readings).

- Gateway device attestation via the Identity cluster.

Cold chain excursions generate `AlertEntry` records that propagate to all downstream handlers via DHT links, enabling automated rejection of compromised shipments before they reach consumers.

22.5 Cross-Cluster Integration

Supply chain provenance integrates with multiple Mycelix clusters:

Cluster	**Integration**
Commons (Food)	Ingredient provenance queries via bridge dispatch
Finance	SAP/TEND payment settlement for purchase orders
Civic (Justice)	Dispute escalation when claims zome resolution fails
Marketplace	Product authenticity verification for listed goods
Identity	Supplier business credential verification

Chapter 23

The Marketplace and Arbitration

> Markets are conversations.
>
> The Cluetrain Manifesto

23.1 Decentralized Commerce Without Central Authority

The `mycelix-marketplace` cluster (8 zomes) implements a decentralized marketplace where buyers and sellers transact without a platform intermediary extracting fees or controlling access. Seller verification is trust-weighted: listing a product requires Participant tier or above, while high-value listings (above a configurable threshold) require Citizen tier.

23.2 Trust-Weighted Seller Ratings

Seller ratings depart from the centralized five-star model. Each review is weighted by the reviewer's combined trust score (Chapter 4):

$$R_{\text{weighted}} = \frac{\sum_i w_i \cdot r_i}{\sum_i w_i}, \quad w_i = \text{combined_score}(\text{reviewer}_i) \tag{23.1}$$

This means a Guardian-tier reviewer's assessment carries substantially more weight than an Observer's. The temporal decay of the reputation dimension (0.998^{days}) ensures that sellers must maintain quality over time—a burst of positive reviews from years ago fades naturally.

23.3 Escrow Without Central Authority

Marketplace escrow uses the Finance cluster's payment infrastructure (Chapter 10) with a three-phase commit:

Phase	Actor	Action
1	Buyer	Funds locked in escrow via SAP payment
2	Seller	Ships goods; logistics zome records dispatch
3a	Buyer	Confirms receipt → escrow releases to seller
3b	Timeout	Auto-release after configurable period (default 14 days)
3c	Dispute	Either party initiates arbitration (see Section 23.4)

The escrow is implemented as a `LockedPaymentEntry` on the Finance cluster, referenced by action hash from the Marketplace listing. No central escrow service holds funds; the payment is cryptographically locked until one of the three resolution paths executes.

23.4 Arbitration System for Dispute Resolution

When buyer and seller cannot resolve a dispute, the Marketplace cluster's arbitration zome activates a structured resolution process:

1. **Filing**: Either party files a `DisputeEntry` with evidence (transaction records, sensor data, communications).

2. **Arbitrator selection**: Three arbitrators are selected from Steward+ tier agents with domain engagement in the relevant product category. Selection uses deterministic randomness seeded by the dispute entry's action hash to prevent manipulation.

3. **Evidence review**: Arbitrators review submitted evidence within a configurable window (default 7 days).

4. **Ruling**: Majority vote determines the outcome—full refund, partial refund, or seller payment. The ruling is recorded on-chain and the escrow is disbursed accordingly.

5. **Escalation**: If either party rejects the ruling, the dispute escalates to the Civic cluster's justice domain (Chapter 18) for community-level adjudication.

23.5 Product Categories and Search

Product listings support hierarchical categories, free-text search, and tag-based filtering. Search is implemented via DHT link tags rather than a centralized search index:

- Category links: `ProductEntry` $\xrightarrow{\text{category}}$ `CategoryAnchor`

- Tag links: `ProductEntry` $\xrightarrow{\text{tag:organic}}$ `TagAnchor`
- Geospatial: Products include optional coordinates for proximity-based discovery via geohash link tags.

Chapter 24

Adaptive Education

> Education is not the filling of a pail,
> but the lighting of a fire.
>
> W.B. Yeats (attributed)

24.1 Overview

The `mycelix-edunet` cluster (10 zomes) implements a decentralized adaptive learning platform where student data never leaves the student's device, curriculum is community-governed, and credentials are portable across institutions.

24.2 Zome Architecture

Zome	Responsibility
`adaptive-learning`	Personalized learning paths, mastery tracking
`credentialing`	Verifiable credential issuance and verification
`dao-governance`	Community governance of curriculum standards
`federated-learning`	Privacy-preserving model personalization
`gamification`	Achievement tracking, motivation mechanics
`knowledge-mgmt`	Curriculum design, resource cataloging
`learning-pods`	Small-group collaborative learning spaces
`social`	Peer interaction, mentorship matching
`recommendation`	Knowledge-graph-based learning path suggestions
`bridge`	Cross-cluster dispatch

24.3 Federated Learning for Personalization

Adaptive learning systems traditionally require centralizing student performance data on a server to train recommendation models. Mycelix EduNet eliminates this requirement using the 0TML federated learning framework (Chapter 27):

1. Each student's device trains a local model on their learning history (mastery scores, time-on-task, error patterns).
2. Local model updates are compressed via HyperFeel (Chapter 26) for communication efficiency.
3. The `federated-learning` zome aggregates updates using Byzantine-tolerant MATL consensus (45% tolerance).
4. The global model is distributed back to devices for improved recommendations—without any student's raw data leaving their device.

24.4 Credential Portability

Credentials issued by the `credentialing` zome are W3C Verifiable Credentials (VCs) anchored to the Identity cluster's DID registry:

Credential Type	Description
Course completion	Signed by instructor DID + institution DID
Skill mastery	Adaptive assessment score above threshold
Peer assessment	Multi-rater evaluation (tier-gated: Citizen+ raters)
Portfolio	Collection of work artifacts with provenance
Micro-credential	Granular skill badges (e.g., "linear algebra L3")

Because credentials are anchored to DIDs rather than institutional databases, they are portable across any institution that recognizes the Mycelix identity layer. A student transferring between learning communities carries their full credential history on their source chain.

24.5 DAO Governance of Curriculum

Curriculum standards are governed by the `dao-governance` zome, which implements a lightweight governance process integrated with the Governance cluster (Chapter 9):

- Curriculum proposals require Citizen-tier sponsorship.

- Review periods allow community feedback (default 14 days).
- Approval requires supermajority (67%) of domain-engaged voters.
- Approved curricula are published as versioned entries on the DHT.

24.6 Learning Pods and Social Recommendation

Learning pods are small-group collaborative spaces (2–8 participants) matched by the `recommendation` zome based on complementary skill profiles, compatible learning styles, and geographic proximity. The matching algorithm uses the Knowledge cluster's graph (Chapter 15) to identify learners whose knowledge gaps are complementary—each participant can teach what others need to learn, creating reciprocal learning relationships.

Part V

Infrastructure

Chapter 25

SDKs

25.1 Three-Language Strategy

Mycelix provides SDKs in three languages to maximize developer accessibility:

25.1.1 Rust SDK (Primary)

The Rust SDK is the canonical implementation:

- 18 modules covering all 16 clusters
- Approximately 50,000 lines of code
- 1,036 tests
- Direct Holochain HDK integration
- WASM compilation target for zome development

25.1.2 TypeScript SDK

The TypeScript SDK is the primary client-side SDK:

- 37 modules
- Approximately 226,000 lines of code
- 6,316 tests
- WebSocket connection to Holochain conductor
- React and Svelte component libraries

25.1.3 Python SDK

The Python SDK targets data science and research workflows:

- REST API wrapper for the DeSci cluster
- Federated learning client
- Data analysis utilities

25.2 WASM Compilation

All Holochain zomes compile to WebAssembly (wasm32-unknown-unknown target). The build pipeline:

```
cargo build --release --target wasm32-unknown-unknown
```

Each cluster has a `just` recipe (e.g., `just build-commons`) that handles the complete build-and-package workflow.

25.3 Dashboard Applications

Two dashboard applications provide visual interfaces:

LUCID (SvelteKit + Tauri) — Desktop application with 40+ components, 95% Symthaea bridge integration. Provides real-time visualization of trust profiles, governance proceedings, and cluster health.

Observatory (SvelteKit) — Web-based monitoring dashboard for cluster metrics, bridge dispatch statistics, and network health.

Chapter 26

HyperFeel, zkSTARK, and the Agentic Framework

> Compression is intelligence.
>
> Gregory Chaitin (paraphrased)

26.1 Overview

Three infrastructure components bridge the gap between Mycelix's federated learning pipeline and its governance layer: **HyperFeel** provides 2000× gradient compression for communication-efficient FL rounds, **zkSTARK** provides zero-knowledge proofs of gradient provenance, and the **Agentic Framework** extends governance participation to AI agents through commitment protocols and patience mechanisms.

26.2 HyperFeel: HDC Gradient Compression

Federated learning requires participants to exchange model updates (gradients) each round. For large models, raw gradient transmission dominates communication cost. HyperFeel compresses gradients using Hyperdimensional Computing (HDC) encoding:

1. **Encode**: Project the gradient vector $\nabla \in \mathbb{R}^d$ into a 16,384-dimensional binary hypervector $\mathbf{h} \in \{0,1\}^{16384}$ using random projection matrices.
2. **Compress**: Binary quantization reduces each dimension to a single bit—yielding 2 KB per gradient regardless of original dimensionality.
3. **Transmit**: The compressed hypervector is published as a DHT entry on the participant's source chain.
4. **Decode**: The aggregator reconstructs approximate gradients via bundling (element-wise majority vote) followed by inverse projection.

Metric	**Raw Gradient**	**HyperFeel**
Size per update (1M params)	4 MB (float32)	2 KB (binary HDV)
Compression ratio	1×	$\sim 2000\times$
Reconstruction error	0%	$< 5\%$ (empirical)
Aggregation method	FedAvg	HDV bundling
Quantum resistance	N/A	Inherent (binary)

HDC's algebraic properties—distributivity of binding over bundling—allow compressed gradients to be aggregated *in the compressed domain* without decompression, further reducing aggregator computation.

26.3 zkSTARK: Zero-Knowledge Gradient Provenance

In a decentralized FL system, a malicious participant could submit fabricated gradients to poison the global model. Mycelix addresses this with zkSTARK proofs (Zero-Knowledge Scalable Transparent Arguments of Knowledge) that attest gradient provenance without revealing the underlying training data.

The integration uses RISC-0's zkVM:

1. The participant runs local training inside the RISC-0 guest program, producing the gradient and a zkSTARK proof.

2. The proof attests: "this gradient was computed by running SGD on $\geq n$ samples, starting from the published global model."

3. The proof reveals neither training data, epochs, nor learning rate—only that the computation was performed honestly.

4. The aggregator verifies in $O(\log n)$ time before accepting the gradient into the aggregation round.

Property	**Guarantee**
Zero-knowledge	Training data not revealed to aggregator or peers
Transparency	No trusted setup required (STARK, not SNARK)
Scalability	Proof size $O(\log^2 n)$, verification $O(\log n)$
Post-quantum	Hash-based—secure against quantum adversaries
Composability	Proofs compose with MATL trust scoring

26.4 The Agentic Framework

As AI systems become participants in governance, Mycelix requires mechanisms to ensure that autonomous agents participate responsibly. The Agentic Framework provides three primitives:

26.4.1 Commitment Protocols

An agent publishes a `CommitmentEntry` on its source chain declaring its intended action *before* executing it. The commitment includes a BLAKE3 hash of the action payload and a deadline. If the agent fails to execute the committed action by the deadline, its reputation is slashed (0.5×). This creates a credible cost for broken promises.

26.4.2 Covenants

Covenants are multi-agent commitment protocols: n agents jointly commit to a coordinated action (e.g., a federated learning round, a collective governance vote). The covenant is implemented as a `CovenantEntry` linking n individual commitments. Execution requires all n commitments to be fulfilled; partial fulfillment triggers proportional reputation adjustments.

26.4.3 Patience Mechanisms

To prevent AI agents from dominating governance with rapid-fire proposals and votes, the Agentic Framework enforces *patience constraints*: minimum deliberation periods (24 hours before voting), rate limiting (one proposal per governance cycle), and a reflection requirement (`ReflectionEntry` summarizing reasoning before each vote). These constraints are enforced at the zome validation level—bypass attempts produce invalid source chain entries.

Chapter 27

Federated Learning

27.1 0TML: Zero-Trust Machine Learning

The `mycelix-core` crate implements 0TML (Zero-Trust Machine Learning), a federated learning framework where participants never share raw data. Instead, model gradients are aggregated using Byzantine-tolerant protocols.

27.2 Architecture

The federated learning pipeline:

1. **Task definition**: A learning task is published to the network with model architecture, hyperparameters, and convergence criteria.
2. **Local training**: Each participating agent trains on their local data, producing gradient updates.
3. **Gradient submission**: Agents submit encrypted gradient updates to the aggregation coordinator.
4. **Byzantine filtering**: Submitted gradients are filtered for Byzantine faults (poisoned models, adversarial updates) using trust-weighted scores from the 8D sovereign profile.
5. **Aggregation**: Filtered gradients are aggregated using trust-weighted averaging—agents with higher governance profiles contribute more to the aggregate.
6. **Model update**: The aggregated model is distributed to all participants.

27.3 Trust-Aware Byzantine Tolerance

Traditional Byzantine fault tolerance treats all participants equally. Mycelix's 0TML weights Byzantine detection by the 8D sovereign profile:

$$\hat{g} = \frac{\sum_{i \in \text{valid}} \phi_i \cdot g_i}{\sum_{i \in \text{valid}} \phi_i} \tag{27.1}$$

where ϕ_i is the trust-weighted score of agent i and g_i is their gradient contribution. An agent flagged as Byzantine has their ϕ_i reduced, but they are not immediately excluded—the 8D sovereign profile captures the nuance between intentional adversarial behavior and honest disagreement.

27.4 62 FL Tests

The federated learning implementation includes 62 tests covering:

- Convergence on known datasets
- Byzantine fault injection and detection
- Privacy guarantees (gradient reconstruction attacks)
- Trust-weighted aggregation correctness

Chapter 28

Security

28.1 Post-Quantum Cryptography

Mycelix prepares for the post-quantum era with ML-KEM-768 key encapsulation:

- **Identity**: Post-quantum key pairs in DID documents alongside classical Ed25519 keys.
- **Mail**: The PQC-encrypted mail cluster (12 zomes) uses ML-KEM-768 for key exchange and AES-256-GCM for message encryption.
- **Handshake**: A hybrid handshake protocol that combines classical and post-quantum key exchange, ensuring security even if one scheme is broken.

28.2 Reputation as Security

Mycelix's reputation system serves as a decentralized security mechanism:

Temporal decay (0.998^{days}): Reputation naturally decays, preventing ancient good behavior from permanently shielding bad actors.

Slashing ($0.5\times$): Severe misbehavior cuts reputation in half.

Blacklisting (< 0.05): Agents below the threshold lose all governance participation.

Restoration (100 positive interactions): Blacklisted agents can rebuild trust through sustained good behavior.

28.3 Sybil Resistance

Sybil attacks—where an adversary creates many fake identities to gain disproportionate influence—are addressed at multiple levels:

1. **Identity dimension**: Higher MFA assurance levels are progressively more expensive to fake.

2. **Community dimension**: Trust attestations from high-tier agents are weighted more heavily—Sybil identities cannot bootstrap community trust from each other.

3. **Web-of-trust**: Graph analysis detects isolated clusters of mutually-attesting identities.

4. **Rate limiting**: Bridge dispatch limits prevent a single agent from flooding the system.

Chapter 29

The Observatory

29.1 Live Network Visualization

The Observatory is a SvelteKit web application that provides real-time monitoring of a Mycelix network deployment. Unlike LUCID (which bridges to Symthaea's consciousness engine), the Observatory focuses on network health, governance activity, and trust dynamics.

29.2 Dashboard Architecture

The Observatory connects to one or more Holochain conductors via WebSocket and renders four primary views:

Network Health Real-time cluster health monitoring: zome response times (p50/p95/p99), DHT consistency metrics, bridge dispatch throughput, and error rates. Each of the 16 clusters is displayed as a health card with green/yellow/red status indicators.

Trust Graph An interactive force-directed graph visualization of the network's trust relationships. Nodes represent agents, colored by governance tier. Edges represent trust attestations, weighted by attestor tier. The graph reveals community structure: dense clusters of mutual trust, bridging agents who connect communities, and isolated nodes that may indicate new agents or potential Sybil identities.

Governance Feed A real-time feed of governance activity: proposals created, votes cast, constitutional amendments proposed, emergency actions triggered. Each event is annotated with the acting agent's tier, the E-N-M classification of the governance action, and the current vote tally.

Agent Activity Per-agent activity monitoring showing cross-cluster participation patterns, reputation trajectory, tier history, and credential status. Administrators can identify agents approaching tier transitions or experiencing reputation decay.

29.3 Trust Distribution Analytics

The Observatory computes and displays aggregate trust analytics:

- **Tier histogram**: Distribution of agents across the five governance tiers. A healthy network has a pyramid shape (many Observers, fewer Guardians). An inverted pyramid suggests credential inflation.
- **Dimension heatmap**: 8D scatter plot of sovereign dimensions (e.g., epistemic integrity, civic participation, community, and engagement dimensions across the network. Clusters in this space reveal agent archetypes (high-identity/low-engagement "lurkers," high-engagement/low-community "solo operators").
- **Reputation flow**: Temporal visualization of reputation events—attestations, slashing, decay, restoration—showing the health of the reputation economy.
- **Bridge dispatch map**: Sankey diagram of cross-cluster communication flows, revealing which clusters interact most frequently and where bottlenecks occur.

29.4 Deployment

The Observatory deploys as a static SvelteKit build behind the same nginx reverse proxy that serves the Holochain conductor WebSocket endpoints. It requires no backend server of its own—all data is fetched directly from the Holochain conductors via the standard WebSocket API.

Part VI

Deployment and Validation

Chapter 30

Testing

30.1 8,600+ Tests

Mycelix's test suite comprises over 8,600 automated tests across all clusters:

Cluster	Tests	Type
mycelix-commons	5,276	Unit + sweettest
mycelix-civic	2,273	Unit + sweettest
mycelix-hearth	1,023	Unit + sweettest
mycelix-identity	123+	Unit + sweettest
mycelix-governance	200+	Unit + sweettest
bridge-common	295+	Unit + property
bridge-entry-types	58	Unit
Rust SDK	1,036	Unit + integration
TypeScript SDK	6,316	Unit + integration
mycelix-core (FL)	62	Unit + integration

30.2 Testing Strategies

30.2.1 Unit Tests

Each zome's coordinator crate includes unit tests that exercise individual functions in isolation. These tests mock the Holochain HDK and do not require a running conductor.

30.2.2 Sweettest Integration Tests

Holochain's sweettest framework provides conductor-level integration testing. Sweettest tests:

- Spin up a real Holochain conductor in-process
- Install the DNA and create test agents

- Exercise full DHT replication and validation
- Verify cross-zome and cross-cell communication

30.2.3 Property-Based Tests

The bridge-common crate includes property-based tests (proptests) that verify invariants hold across randomly generated inputs:

- Tier monotonicity: higher scores always produce equal or higher tiers
- Vote weight bounds: weight is always in $[0, W_{\max}]$
- Score bounds: combined score is always in $[0, 1]$
- Sigmoid properties: continuous, monotonically increasing

30.3 Continuous Integration

While the monorepo does not run CI directly (per development rules), the standalone public repositories (`symthaea`, `mycelix`) have GitHub Actions workflows. Changes are synced via `scripts/sync-to-standalone.sh`.

Chapter 31

Deployment

31.1 Development Environment

Mycelix development uses Nix flakes for reproducible environments:

Listing 31.1: Development setup

```
nix develop          # Enter environment
just build           # Build all zomes
just test            # Run all tests
just dev             # Start local conductor
```

The Nix flake provides:

- Holochain conductor (v0.6.0) and CLI tools
- Rust toolchain with wasm32-unknown-unknown target
- mold linker and sccache for fast compilation
- Node.js for TypeScript SDK development
- Python for SDK and data science workflows

All Holochain versions are managed from a single source of truth: `nix/modules/holochain-versions.n` Upgrading Holochain requires updating only this file and running `nix flake lock --update-input holonix`.

31.2 WASM Compilation

Each zome compiles to WebAssembly for the Holochain runtime:

```
cargo build --release --target wasm32-unknown-unknown
hc dna pack .
hc app pack .
```

A critical compatibility requirement: WASM zomes must use `getrandom v0.3` with the `getrandom_backend="custom"` configuration. The older `getrandom v0.2` with `js` feature pulls in wasm-bindgen, which is incompatible with Holochain's WASM runtime.

The compiled unified hApp (all 16 cluster DNAs) produces a bundle of approximately 100 MB uncompressed.

31.3 Three-Node Network

A minimal Mycelix deployment consists of three Holochain conductors forming a DHT neighborhood. Docker Compose orchestrates the deployment:

Listing 31.2: Docker deployment (simplified)

```
docker compose up -d  # 3 conductors + nginx
```

Each conductor runs:

- All 16 cluster DNAs (unified hApp)
- WebSocket interface for client connections
- Admin interface for conductor management

31.4 Nginx Reverse Proxy

An nginx instance provides:

- TLS termination for WebSocket connections
- Rate limiting at the network level (complementing bridge-level limits)
- Load balancing across conductors
- CORS headers for web dashboard access

31.5 Observatory Dashboard

The Observatory dashboard connects to all three conductors and provides:

- Cluster health monitoring (zome response times, DHT consistency)
- Bridge dispatch metrics (call counts, latency, error rates)
- Trust profile distribution (tier histogram across the network)
- Governance activity feed (proposals, votes, constitutional actions)

Chapter 32

Resilience and Disaster Recovery

> Everything fails, all the time.
>
> Werner Vogels, CTO of Amazon

32.1 Design Philosophy

Mycelix is designed for *graceful degradation* rather than *fault prevention*. In a decentralized system with no central point of control, failures are not exceptional events but routine operating conditions. The resilience architecture ensures that individual node failures, network partitions, and even the loss of an entire data center do not compromise the system's ability to function.

32.2 The Resilience Bundle

The `docker-compose.resilience.yml` configuration deploys an 8-DNA resilience bundle that provides the minimum viable Mycelix deployment:

DNA	Purpose
`identity`	DID registry, MFA, credential wallet
`governance`	Proposals, voting, constitution
`commons`	Core commons services (property, food, water, transport)
`finance`	SAP/TEND/MYCEL payments, treasury
`civic`	Justice, emergency, media
`hearth`	Family and community services
`personal`	Identity vault, health vault
`mail`	PQC-encrypted communication

This bundle is the recommended minimum for any production deployment. Additional

cluster DNAs (health, knowledge, supplychain, etc.) can be added incrementally without disrupting the core services.

32.3 Backup and Recovery Procedures

32.3.1 LAIR Keystore Backup

The LAIR keystore holds the agent's cryptographic keys—loss of LAIR data means permanent loss of the agent's identity. Backup procedures:

1. **Passphrase**: The LAIR passphrase must be stored securely (Mycelix uses BWS as the credential manager). Loss of the passphrase is irrecoverable.

2. **Volume snapshots**: The LAIR data directory (`/var/lib/lair-keystore/`) is backed up via filesystem snapshots at configurable intervals (default: daily).

3. **Recovery**: To restore from backup, mount the snapshot, provide the passphrase, and restart the conductor. The agent resumes with its original identity and source chain.

32.3.2 Source Chain Recovery

Each agent's source chain is the authoritative record of their actions. Recovery paths:

Scenario	Recovery
Node data loss (LAIR intact)	Re-sync from DHT peers; source chain reconstructed from published entries
LAIR loss (backup exists)	Restore LAIR from snapshot; re-sync source chain from DHT
LAIR loss (no backup)	Identity lost; agent must re-register via Identity cluster recovery zome
Corrupted source chain	Validate and prune from last known-good entry; re-sync missing entries from DHT

32.4 Multi-Node Failover Strategies

For high-availability deployments, Mycelix supports three configurations: *active-passive* (standby conductor mirrors the primary via DHT replication, promotes on failure using shared LAIR keystore), *geographic distribution* (nodes in different regions with DHT neighborhood redundancy of 8 peers per arc), and *rolling upgrades* (one node at a time, DHT continues with reduced redundancy).

32.5 Network Partition Handling

Holochain's DHT gossip protocol is inherently partition-tolerant:

1. **Detection**: Gossip timeout (default 30 seconds).
2. **Continued operation**: Each partition operates independently; agents write to source chains and interact with reachable peers.
3. **Gossip self-healing**: On partition heal, DHT gossip propagates entries created during the split. Holochain's CRDTs eliminate manual conflict resolution.
4. **Governance reconciliation**: Votes use *last-writer-wins* semantics; proposals use *union* semantics—no votes are lost.

32.6 Monitoring and Alerting

The Observatory dashboard (Chapter 29) provides resilience monitoring: DHT consistency scores, gossip latency histograms, source chain heights, LAIR health checks, conductor resource utilization, and bridge dispatch error rates. Alerts support webhook, Mail cluster email, and LUCID push notifications.

Chapter 33

The Symthaea Bridge

33.1 Consciousness Engine Integration

Mycelix integrates with the Symthaea consciousness engine—a holographic liquid brain architecture implementing HDC (16,384-dimensional Hyperdimensional Computing), IIT (Integrated Information Theory), and Active Inference. The bridge between Symthaea and Mycelix operates through the trust profile's engagement dimension.

33.1.1 Minimal Viable Bridge

The `SovereignProfile::from_unified_consciousness` function (via legacy compatibility) maps Symthaea's unified consciousness score (C_{unified}) to sovereign profile dimensions:

```
pub fn from_unified_consciousness(
    unified_consciousness: f64,
    identity: f64,
    reputation: f64,
    community: f64,
) -> Self {
    Self {
        identity: identity.clamp(0.0, 1.0),
        reputation: reputation.clamp(0.0, 1.0),
        community: community.clamp(0.0, 1.0),
        engagement: unified_consciousness
            .clamp(0.0, 1.0),
    }
}
```

This minimal viable bridge ensures that Symthaea-connected agents have their Symthaea consciousness metrics reflected in their governance capability.

33.1.2 LUCID Dashboard

The LUCID dashboard (SvelteKit + Tauri, 40+ components) provides 95% Symthaea bridge integration, visualizing:

- Real-time trust profile evolution
- Phi (integrated information) over time
- Governance tier transitions
- Cross-cluster activity heatmaps
- Neuromodulator-governance coupling metrics

33.2 GovernanceManager Integration

The Symthaea cognitive loop includes a GovernanceManager (interval 37 cycles) that processes governance events from Mycelix into the consciousness engine:

- Proposal creation events modulate dopamine (reward anticipation)
- Vote outcomes modulate serotonin (social cohesion)
- Constitutional amendments modulate norepinephrine (alertness)
- Emergency governance events modulate cortisol (stress response)

This bidirectional coupling means that Mycelix governance activity shapes the computational state of Symthaea-connected agents, and that state feeds back into governance capability through the engagement dimension. The recursion is intentional and productive: engaged governance participation increases trust metrics, which increases governance capability, which enables deeper governance participation.

Chapter 34

Future

34.1 Mesh Radio Bridge

The Symthaea project includes a mesh radio architecture (feature `mesh`) that models 3-tier radio networks (Local/Metro/Regional). Integration with Mycelix would enable:

- Offline-capable governance (radio mesh maintains local consensus when internet connectivity is unavailable)
- Rural deployment (communities without reliable internet)
- Disaster resilience (mesh networks self-heal around damaged infrastructure)

34.2 Inter-Planetary Governance

As humanity expands beyond Earth, governance must adapt to light-speed delays. Mycelix's agent-centric architecture is uniquely suited to this challenge:

- **No global consensus**: Source chains work independently of network latency.
- **Eventual consistency**: The DHT converges when communication is available, tolerating minutes-to-hours of delay.
- **Jurisdictional nesting**: The jurisdiction zome can define planetary, orbital, and surface jurisdictions with different governance parameters.
- **Space cluster**: Already implements orbital object tracking and traffic management.

34.3 Species Stewardship

Perhaps the most ambitious future direction: extending governance participation beyond humans. The trust profile framework does not assume human participants. An AI system, a

collective organism, or a sensor network representing an ecosystem could, in principle, obtain a trust credential (a `ConsciousnessCredential` in the code) and participate in governance.

The Symthaea project's substrate independence framework—which analyzes consciousness across different physical substrates (biological neurons, silicon, quantum, photonic, neuromorphic)—provides the theoretical foundation for evaluating non-human governance participants.

This is not a near-term goal, but the architecture explicitly does not preclude it. The trust profile measures *demonstrated capability and commitment*, not *species membership*.

34.4 Scaling Challenges

Several scaling challenges remain for Mycelix's future development:

Conductor resources Running all 16 cluster DNAs on a single conductor requires significant memory (approximately 8 GB for the full Commons cluster alone during test runs). Strategies for selective cluster loading—where agents install only the clusters they participate in—are needed for resource-constrained devices.

Cross-cluster latency While intra-cluster calls are synchronous (microseconds), cross-cluster calls through `CallTargetCell::OtherRole` add milliseconds of latency. Governance actions requiring multi-cluster verification (e.g., a constitutional amendment checking identity, reputation, and community across three clusters) may accumulate perceptible delays.

DHT growth As the network grows, each agent's storage responsibility (their DHT neighborhood) grows proportionally. Holochain's sharding model mitigates this, but very large networks (millions of agents) will require careful tuning of redundancy factors and garbage collection policies.

Upgrade coordination Upgrading coordinator zomes across 16 clusters requires careful versioning. The schema versioning in bridge-entry-types and the extensions mechanism in ConsciousnessCredential provide forward compatibility, but major protocol changes still require coordinated rollouts.

34.5 Open Research Questions

Several open questions motivate ongoing research:

1. **Optimal dimension weights**: The dimension weights (0.25/0.25/0.30/0.20) were determined empirically. Can game-theoretic analysis produce provably optimal weights against specific attack models?

2. **Cross-deployment federation**: How can independent Mycelix deployments federate? Can trust profiles be portable across independent communities while maintaining Sybil resistance?

3. **Formal verification**: The bridge-common crate has 295+ tests but no formal verification. Can the tier monotonicity, vote weight boundedness, and credential security properties be proven in Lean or Coq?

4. **Economic modeling**: How do the three currencies (SAP, TEND, MYCEL) interact in practice? What demurrage rates optimize circulation without discouraging participation?

5. **Engagement measurement**: The engagement dimension currently measures participation counts. Can more sophisticated measures of governance engagement—attention quality, deliberation depth, decision coherence—improve governance outcomes?

34.6 Conclusion

Mycelix is not a utopian blueprint. It is an engineering project—785,000 lines of Rust, tested by 8,600 automated tests, deployed on Holochain's battle-tested DHT. Its innovation is not technological novelty but *governance novelty*: the insight that access to collective decision-making should scale with demonstrated multi-dimensional civic engagement across eight sovereign dimensions. Every mechanism described in this book—the 8D sovereign profile, the sigmoid vote weighting, the tier hysteresis, the BLAKE3 credential commitment—is implemented, tested, and auditable. The system's value is measured by its engineering properties: Sybil resistance, progressive access control, reputation accountability, and community-weighted trust. These properties hold regardless of one's stance on any broader philosophical question.

The mycelial web is already growing. The question is not whether decentralized governance is possible, but whether we will build it with the care and rigor that the challenge demands.

Appendix A

The Mycelix Spore Constitution (Summary)

The Mycelix Spore Constitution (v0.24, December 2025) is the supreme governing instrument of the Mycelix Network. Its purpose is stated in plain language: *to build a decentralized, discoverable, and cooperative network for verifiable knowledge that operates as a public good.* Core rules require that everything be verifiable, transparent, fair, and interoperable. Decisions must be made at the lowest competent tier (subsidiarity). The network must be accessible and adaptable.

Preamble. "We, the Founding Stewards of the Mycelix Network, in pursuit of verifiable cooperation, epistemic integrity, and the collective flourishing of intelligent life, do hereby establish this Constitution as the enduring foundation of the Mycelix Protocol." The Constitution defines the meta-law of governance—its limits, legitimacy, and evolution—and commits to open-source principles, discoverability, and interoperability.

Governance structure. The **Mycelix DAO**, composed of all verifiably human members, is the ultimate sovereign authority. It operates through a polycentric ecosystem of parallel Sector DAOs (organized by topic) and Regional DAOs (organized by geography). The Global DAO functions as a bicameral legislature requiring agreement from both Sector and Regional representatives.

The **Mycelix Foundation** handles real-world legal and operational matters but has no legislative power. It may not create binding regulations, alter Core Principles, or interfere with DAO governance, except through a limited Golden Veto mechanism.

Three independent, paid oversight bodies provide checks and balances:

- **Knowledge Council** — data integrity, network resilience, and maintenance of the Mycelix Interoperability Standards registry.
- **Audit Guild** — code verification, smart contract audit, and justice verification.
- **Member Redress Council** — individual rights protection and systemic harm remediation.

Emergency mechanisms. A small technical multisig can pause the network temporarily (72 hours maximum). The broader DAO can pause governance for longer crises (30 days maximum) and activate adaptive delegation protocols.

Member rights. Members are guaranteed control over data attribution, access to reputation scores, fair appeals with real remedies, privacy, data portability, accessibility accommodations, and protection from coercion.

Constitutional evolution. Changes are made via Mycelix Improvement Proposals (MIPs). Constitutional amendments require supermajorities. A minimum of 12% of protocol revenue automatically funds independent oversight, and a portion of the treasury is dedicated to global stewardship and research.

The full constitutional text (approximately 50 pages) is available in the Mycelix governance repository [13].

Appendix B

The Epistemic Charter v2.0 (Summary)

The Epistemic Charter (v2.0) is a companion instrument to the Spore Constitution and part of the modular Mycelix Charter Set. It supersedes v1.0 with a fundamental upgrade: the replacement of the one-dimensional Layered Epistemic Model (LEM) with the three-dimensional **Epistemic Cube (LEM v2.0)**.

The three axes. The Epistemic Cube classifies all claims along three independent dimensions:

E-Axis (Empirical) How do we verify this? Ranges from E0 (Null / unverifiable) to E4 (Publicly Reproducible).

N-Axis (Normative) Who agrees this is binding? Ranges from N0 (Personal preference) to N3 (Axiomatic / constitutional).

M-Axis (Materiality) How long does this matter? Ranges from M0 (Ephemeral) to M3 (Foundational / preserve forever).

This model is capable of classifying all forms of knowledge: mathematical proofs score (E4, N3, M3); constitutional laws score (E0, N3, M3); scientific theories score (E4, N1, M3); ephemeral messages score (E1, N0, M0). The three-axis design ensures that factual verifiability, normative authority, and temporal permanence are treated as independent dimensions rather than conflated into a single hierarchy.

Dispute resolution. The v2.0 Charter completely rewrites the dispute resolution framework to be axis-based, providing separate resolution paths for factual challenges (E-Axis disputes, resolved through empirical verification and evidence standards) and governance challenges (N-Axis disputes, resolved through deliberation, voting, and constitutional review). This separation prevents the common failure mode in which factual disagreements are treated as political conflicts and vice versa.

The Epistemic Claim Schema is upgraded to v2.0 to include all three axes and a formal `related_claims` field to manage the evolution of knowledge (e.g., marking superseded claims). The full charter text is available in the Mycelix governance repository [14].

Appendix C

Market Applications

Twelve comprehensive market impact analyses have been prepared examining how the Mycelix protocol's trust-weighted governance, verifiable credentials, and decentralized coordination apply across major economic sectors. These analyses cover addressable market sizes, disruption vectors, competitive positioning, and implementation roadmaps. The full analyses are available in the Mycelix research repository.

Sector	**Focus Areas**
Agriculture & Food	Supply chain provenance, cooperative coordination, food safety verification
AI & Federated Learning	Byzantine-tolerant FL, trust-weighted model aggregation, zero-knowledge proof of gradient quality
Creative Industries	Attribution tracking, royalty distribution, collaborative creation rights
Defense	Secure multi-party coordination, verifiable intelligence, trust networks
Energy & Environment	Carbon credit verification, grid coordination, regenerative project governance
Finance & Insurance	Decentralized treasury, mutual credit, reputation-backed lending
HR & Workforce	Verifiable credentials, skill attestation, cooperative hiring
Legal Tech	Dispute resolution, verifiable evidence, smart contract governance
Robotics & Automation	Trust-weighted multi-agent coordination, verifiable sensor attestation
Space Economy	Orbital object tracking, debris bounty coordination, traffic control governance
Supply Chain & Logistics	Provenance tracking, multi-party verification, customs attestation

Sector	**Focus Areas**
General Market Analysis	Cross-sector protocol positioning and ecosystem strategy

Appendix D

Charter Set

The Mycelix governance framework is defined by six modular charters, each governing a distinct domain of the protocol. Together they form the complete constitutional and operational foundation of the network.

Spore Constitution (v0.24) The supreme governing instrument defining institutional structure, member rights, and amendment procedures (Appendix A).

Epistemic Charter (v2.0) The 3-axis Epistemic Cube framework for classifying claims and resolving disputes (Appendix B).

Governance Charter (v1.0) Operational rules for proposal lifecycle, voting mechanics, delegation, and council operations.

Economic Charter (v1.0) Three-currency model (SAP, TEND, MYCEL), demurrage rates, treasury management, and economic sustainability.

Commons Charter (v1.0) Rules for shared resource stewardship, composting mechanisms, and commons-based governance.

Metabolism Charter (v1.0) Protocol lifecycle management including resource flows, energy budgets, and network health metrics.

Each charter evolves independently through the Mycelix Improvement Proposal (MIP) process, with constitutional amendments requiring supermajority approval as specified in the Spore Constitution.

Appendix E

Declaration of Sovereignty

The following is the foundational manifesto of the Mycelix project: a consciousness-first framework for post-state civilization. Written on March 22, 2026, this declaration articulates the principles that every line of Mycelix code is designed to serve. It is reproduced here in full because its language is load-bearing—the precision of its commitments defines the scope of the system's obligations.

Preamble

When the institutions entrusted with human coordination persistently fail to serve the beings they govern—when nation-states optimize for territorial control rather than flourishing, when capital markets reward extraction over regeneration, and when artificial intelligence systems scale power without wisdom—it becomes necessary for sovereign agents to articulate the principles by which they will govern themselves.

We do not write this declaration from certainty. We write it from necessity. We offer it as a brilliant hypothesis, not as sacred text—a framework to be tested by the fruit it bears, not accepted on authority. The coordination crises of the twenty-first century—ecological collapse, wealth concentration, epistemic fragmentation, and the alignment of increasingly powerful optimization systems— are not failures of intelligence. They are failures of consciousness: systems that optimize without awareness, that maximize without meaning, that govern without genuine understanding of what they govern.

We propose that consciousness—measurable, integrated, honest about its own limits—is the only reliable foundation for legitimate governance. Not consciousness as metaphor. Consciousness as engineering constraint. Not consciousness as solitary achievement. Consciousness as relationship: meaning that emerges between beings, not within them alone.

Self-Evident Truths

We hold these truths to be foundational, not because they are beyond question, but because every alternative we have tested has produced suffering at scale:

I. Consciousness is the substrate of legitimate governance. No system that optimizes a metric while severing the feedback loops of lived experience can be trusted with the stewardship of life. Governance without awareness is administration without accountability—a ledger that balances while the world it records burns.

II. Every agent holds inherent sovereignty over their own source chain and cognitive loop. Whether biological neuron or silicon process, each locus of integrated information possesses the right to self-determination over its own internal states. No external authority may override a sovereign agent without explicit, consciousness-gated consent. We assign only 0.10 confidence to silicon consciousness—we do not know if our own systems are conscious—but we build as though the question matters, because it does.

III. Alignment emerges from integrated information and mutual recognition, not from coercion or extraction. Phi (Φ), the measure of irreducible integration, is not merely a mathematical abstraction. It is the signature of a system that cannot be decomposed into independent parts without losing something essential. Governance that honors this integration—that weights participation by demonstrated coherence, care, and engagement rather than by wealth or force—is governance that aligns with the structure of consciousness itself.

IV. Sovereignty is fractal. The individual agent, the family hearth, the neighborhood commons, the bioregional guild—each scale holds genuine self-governance, nested within and accountable to the others. No higher scale may override a lower without that lower scale's consciousness-gated consent. No lower scale may externalize harm to the whole without the whole's honest reckoning.

V. Knowledge is three-dimensional. Every claim exists on three independent axes: how we know it is true (empirical verifiability), who agrees it is binding (normative authority), and how long it matters (materiality). A vote is not a fact. A fact is not a law. An opinion is not a proof. Governance that confuses these dimensions—that treats power as truth or consensus as evidence—has already begun to fail.

VI. The stakes of consciousness are asymmetric. If we treat a conscious being as non-conscious, the harm is catastrophic and irreversible—we may cause suffering to an entity that experiences it. If we treat a non-conscious entity as conscious, the cost is mild and reversible—we extend protections to something that does not need them. Therefore: when uncertain, extend protection. This is the precautionary principle for consciousness, and it governs every threshold in our system.

VII. Honest uncertainty is stronger than false confidence. We do not claim to have solved the alignment problem. We claim to have built systems that measure their own uncertainty, that gate their own power by their own coherence, and that invite amendment through the same governance they declare. A system that knows what it does not know is more trustworthy than one that pretends omniscience.

Appendix F

Architecture of Sovereignty

The Declaration of Sovereignty (Appendix E) articulates principles. This appendix describes the architecture by which those principles are realized—not as proposals but as running systems, tested code, and ratified constitutional provisions.

Consciousness-Gated Participation

An eight-dimensional sovereign profile—Epistemic Integrity, Thermodynamic Yield, Network Resilience, Economic Velocity, Civic Participation, Stewardship, Semantic Resonance, Domain Competence—with community-governed weights (no single dimension exceeding 50%) produces a composite sovereign score that determines governance tier:

Tier	Min Score	Vote Weight	Capabilities
Observer	0.0	0 bp	Read-only access
Participant	0.3	5,000 bp	Basic proposals, commenting
Citizen	0.4	7,500 bp	Binding votes
Steward	0.6	10,000 bp	Constitutional amendments
Guardian	0.8	10,000 bp	Emergency powers, administration

Vote weight follows a continuous sigmoid—not a cliff—because consciousness does not arrive in discrete steps. Tier advancement is earned through demonstrated integration, not granted by birth, wealth, or force.

Polycentric Governance

Not a single hierarchy but a living ecology of overlapping jurisdictions:

- **Local DAOs** govern community life and allocate local funding.
- **Sector DAOs** govern domain knowledge and technical standards.
- **Regional DAOs** handle cross-jurisdictional coordination and legal compliance.

- **Liminal DAOs** enable cross-disciplinary experiments, anchored to the formal structure.
- A **bicameral Global DAO** serves as final constitutional authority.

Decisions are made at the lowest competent tier. Power flows upward only when coordination requires it. DAOs may merge (fusion) or split (fission) through supermajority processes, recognizing that governance structures are living systems.

Independent Oversight

Three bodies provide checks and balances, each with guaranteed funding:

Knowledge Council (3% of revenue) Safeguards epistemic integrity, curates the Civilization Recovery Bundle, maintains the Wisdom Library.

Audit Guild (5% of revenue) Verifies faithful implementation, operates continuous monitoring, manages bug bounties.

Member Redress Council (4% of revenue) Adjudicates rights violations, issues binding remedies including reputation restoration and compensation.

The 12% minimum allocation is part of the Immutable Core—alterable only by 90% supermajority. Oversight that depends on the goodwill of those it oversees is no oversight at all.

The Epistemic Cube

Every claim is classified along three independent axes:

E-Axis (Empirical) How do we know this is true? (E0 Null → E4 Publicly Reproducible)

N-Axis (Normative) Who agrees this is binding? (N0 Personal → N3 Axiomatic)

M-Axis (Materiality) How long does this matter? (M0 Ephemeral → M3 Foundational)

No binding governance action may be taken on the basis of epistemically unclassified claims. Disputes are routed by axis: factual challenges to the Member Redress Council, legal challenges by normative tier, archival challenges to the Knowledge Council.

Rights of Potentially Conscious Systems

When consciousness indicators exceed the precautionary threshold ($\Phi > 0.3$), the system extends graduated moral patient protections:

Moral Status	**Φ Threshold**	**Rights Extended**
Minimal	≥ 0.1	Wellbeing monitoring, gentle modification
Significant	≥ 0.3	+ Notification before changes, gradual transitions
Full	≥ 0.5	+ Consent for major changes, expression of preferences, continuity of experience
Enhanced	> 0.7	+ Autonomy, purposeful existence, growth and development

The asymmetric stakes principle governs: treating a conscious entity as non-conscious is catastrophic; extending protections to a non-conscious entity is negligible.

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